Long range QKD with time and frequency multiplexing in broadband solid state memories

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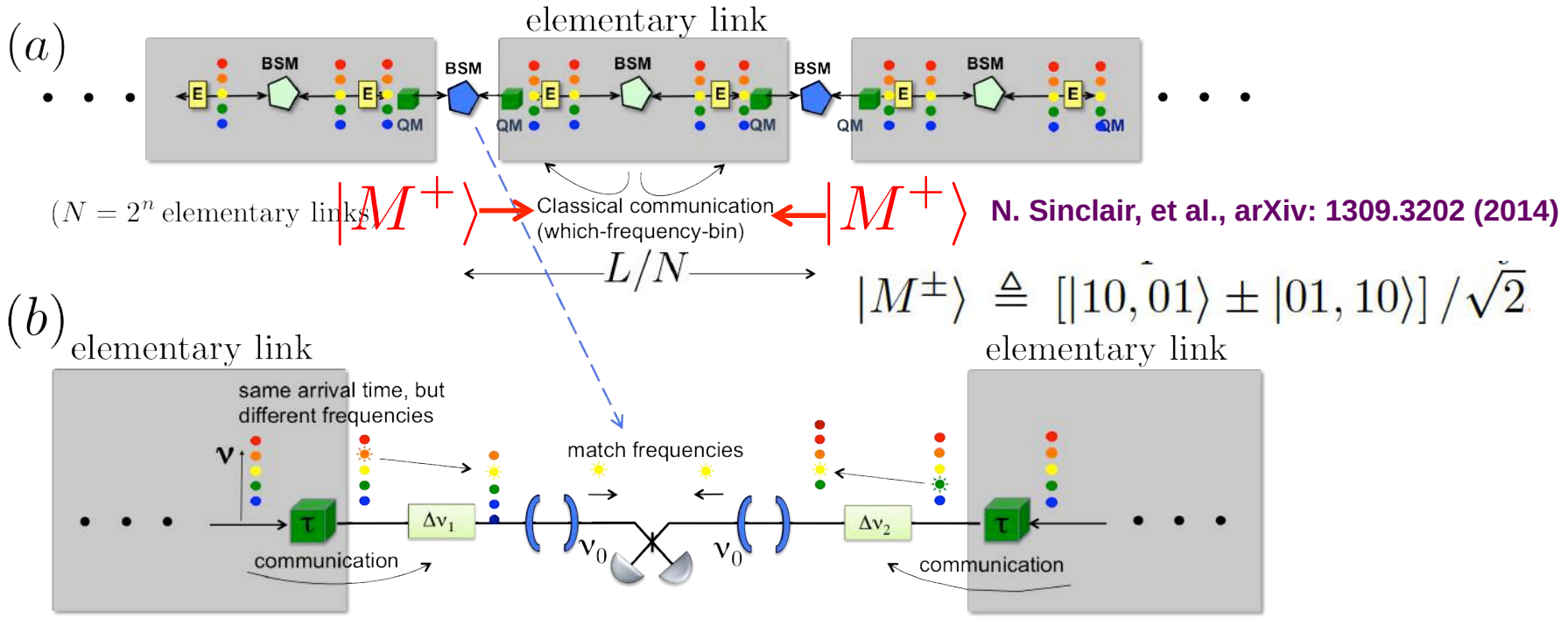
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- Description of the repeater architecture with frequency multiplexing and Bell state measurements
- State propagation with entanglement swapping
- Secret key rate calculation
- Performance of secret key rates with dark clicks in the detectors
- Performance with source imperfections
- Conclusions

QKD with concatenated repeater chain

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- Repeater architecture: connecting *elementary links*



N. Sinclair, et al., arXiv: 1309.3202 (2014)

  - No purification done at intermediate swap levels
  - Linear-optic BSM at memories; could be replaced by more advanced schemes that use non-linearities

$(\eta_e, P_e), (\eta_r, P_r), (\eta_d, P_d)$ Det. Eff. and dark-prob. of elem link, repeater node, and Alice-Bob detectors

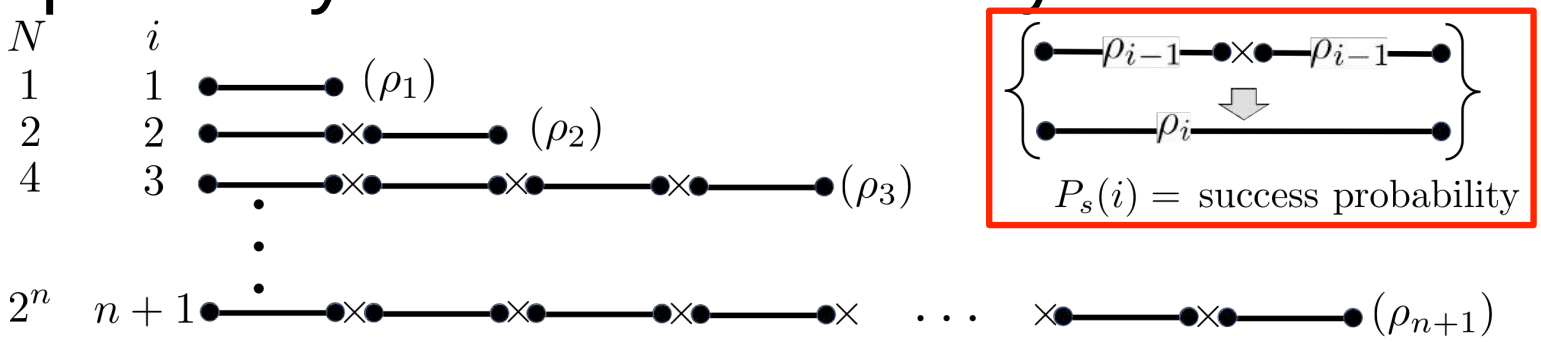
$\lambda_m =$ Sub-unity transmittance for loss at memory, $M =$ No. of freq. modes, $T_q =$ Pulse-slot (qubit) duration

$\lambda =$ Fiber transmittance associated with L/2N km (half of an elem link), $\alpha =$ Fiber loss in dB/km

Connecting elementary links

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- Sequentially connect elementary links…



$$\rho_i = \frac{1}{s_i}\left[a_i|M^+\rangle\langle M^+| + b_i|M^-\rangle\langle M^-| + c_i|\psi_0\rangle\langle\psi_0| + d_i|\psi_1\rangle\langle\psi_1| + d_i|\psi_2\rangle\langle\psi_2| + c_i|\psi_3\rangle\langle\psi_3|\right]$$

$$|\psi_0\rangle = |01,01\rangle, |\psi_1\rangle = |01,10\rangle, |\psi_2\rangle = |10,01\rangle, |\psi_3\rangle = |10,10\rangle, |M^{\pm}\rangle = \left[|\psi_2\rangle \pm |\psi_1\rangle\right]/\sqrt{2}$$

$$s_1 = \underbrace{a_e + b_e}_{\text{Prob. amplitude in Bell-states}} + \underbrace{2c_e}_{\text{Prob. Amplitude in classical correlations}} \quad \Rightarrow \quad P_s(1) = 1 - (1 - 4s_1)^M$$

Probability of successfully creating one elementary link

$$s_i = a_i + b_i + 2(c_i + d_i) = s = a + b + 2c; P_s(i) = 4s,\ 2 \le i \le n+1$$

$$P_{\text{succ}} = \left[P_s(n+1)\right]\left[P_s(n)\right]^2 \ldots \left[P_s(2)\right]^{2^{n-1}}\left[P_s(1)\right]^{2^n} = \frac{1}{4s}\left[4s\left(1 - (1 - 4s_e)^M\right)\right]^N$$

Success probability of connecting two good copies of ρ_{i-1} (to create one good copy of ρ_i), is the same for all i=2, …, n+1. Success probability for ρ_1, the *elementary link,* is higher due to many (M) frequency modes. At each swap stage,

Quantum state propagation

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- Explicit calculation of the state after n swap levels

$$a_i = \frac{1}{2}\left[1 + \left(\frac{a-b}{a+b}\right)^{i-1}\left(\frac{a_e-b_e}{a_e+b_e}\right)\right] z_i,$$

$$b_i = \frac{1}{2}\left[1 - \left(\frac{a-b}{a+b}\right)^{i-1}\left(\frac{a_e-b_e}{a_e+b_e}\right)\right] z_i,$$

$$c_i = \frac{s_i}{4}\left[1 - \frac{z_i}{s_i(1-2w_r)}\left[(1-2w_r)(1-2w_1)\right]^{2^{i-1}}\right],$$

$$d_i = \frac{s_i}{4} - \frac{z_i}{2}\left[1 - \frac{1}{2(1-2w_r)}\left[(1-2w_r)(1-2w_1)\right]^{2^{i-1}}\right]$$

Solving for the exact state required solving the following new variant of the ***logistic map*** from chaos theory:

$$w_{i+1} = w_r + 2(1-2w_r)w_i(1-w_i)$$

These ratios (probability spillovers in classical correlation to that in Bell states), play an important role in the analysis

$$\left\{\begin{array}{c} 2w_1 = \dfrac{2c_e}{a_e+b_e} \\ 2w_r = \dfrac{2c}{a+b} \end{array}\right\}$$

$$z_1 = a_e + b_e,\ z_i = \left(\frac{s^2}{a+b}\right)\left(\frac{1}{(1+2w_1)(1+2w_r)}\right)^{2^{i-1}},\ i \geq 2$$

(a, b, c) functions of: η_r, P_r, λ_m — Repeater node SPDs

(a_e, b_e, c_e) functions of: η_e, P_e, λ — Elementary link SPDs

First explicit analysis of quantum state propagation through repeater chain with detector noise: Useful to analyze other repeater protocols and other applications that use entanglement distribution

Secret-key-rate calculation

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- Secret-key rate, $R_s = \frac{P_1 P_{\text{succ}} R(Q_{n+1})}{2T_q}$ Ferenczi and Lutkenhaus, PRA 85, 052310 (2012)

  $(R(q) = 1 - 2h_2(q))$ Dutton, Fuchs, Guha, Krovi, ArXiv:1404.7183 (2014)
- Sift probability (P_1), success probability (P_{succ}), and QBER (Q_i, i=1,…,n+1) explicitly calculated
- QBER propagation law: $(1 - 2Q_{i+1}) = \frac{t_r}{t_d}(1 - 2Q_i)^2, 1 \leq i \leq n$
- QBER: $Q_i = \frac{1}{2}\left[1 - \frac{t_d}{t_r}(t_r t_e)^{2^{i-1}}\right], 1 \leq i \leq n+1$

$$t_r = \frac{1-2w_r}{1+2w_r} \quad t_e = \frac{1-2w_1}{1+2w_1} \quad t_d = \left(\frac{q_1 - q_2}{q_1 + q_2}\right)^2$$

Detector-noise parameters for 3 sets of SPDs in the system

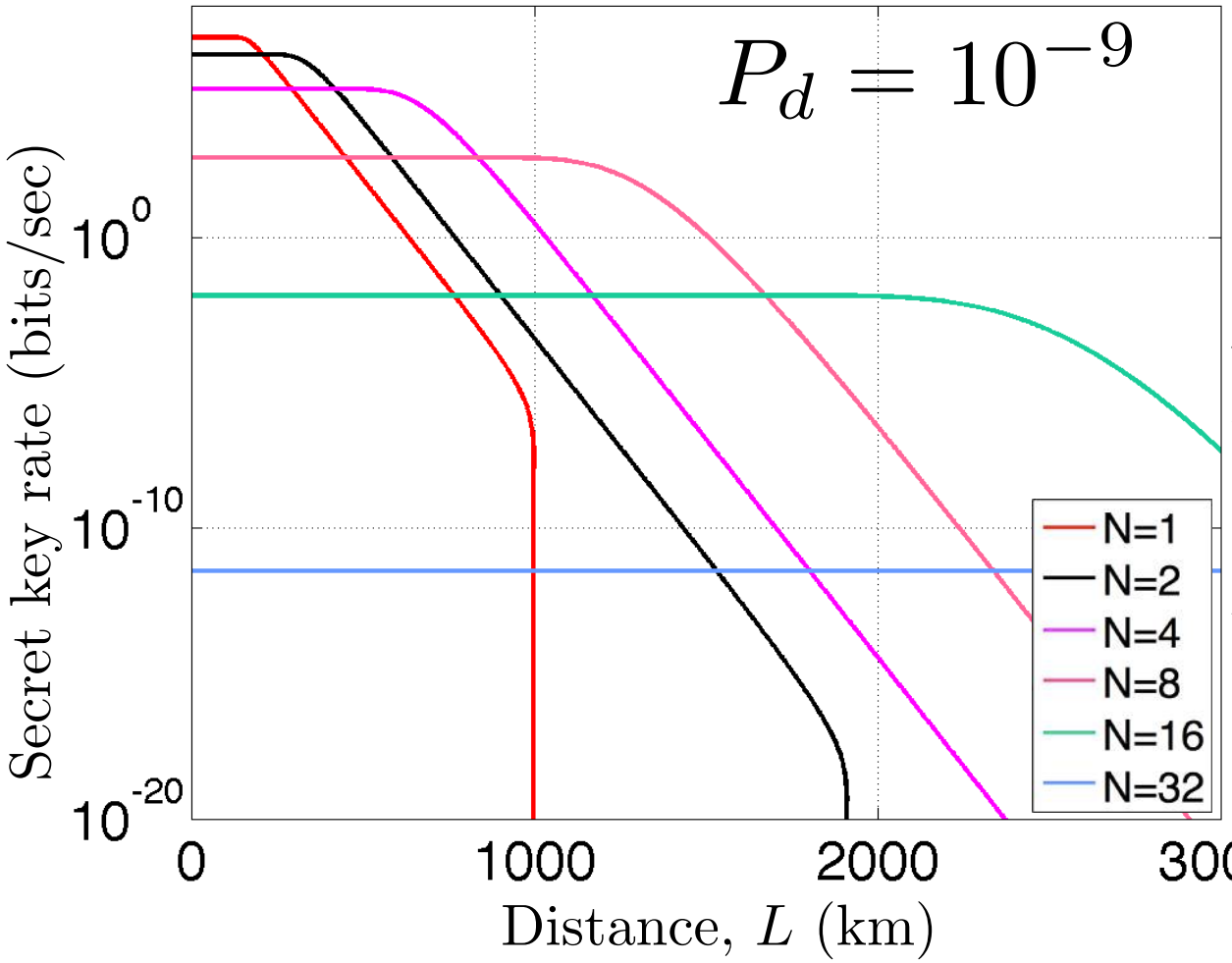
$$q_1 = (1-P_d)A_d,\ q_2 = (1-A_d)P_d,\ \textit{with}\ A_d = \eta_d P_d + 1 - \eta_d$$

Detector noise parameters. e: elementary link detectors; r: repeater (memory) node detectors; d: Alice and Bob's detectors. The t's become 1 if dark click probability of the respective detectors is 0.

Effect of dark-click probability on the rate-distance performance

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- Assume source has zero two-photon emissions



$$\eta_r = \eta_d = \eta_e = 0.9$$

$$T_q = 50\,\text{ns}$$

$$\alpha = 0.15\,\text{dB/km}$$

$$\lambda_m = 10^{-0.1\gamma_m}, \gamma = 1\,\text{dB}$$

$$M = 1000\,\text{frequency modes}$$

$$P_d = P_r = P_e$$

$N =$ number of elementary links

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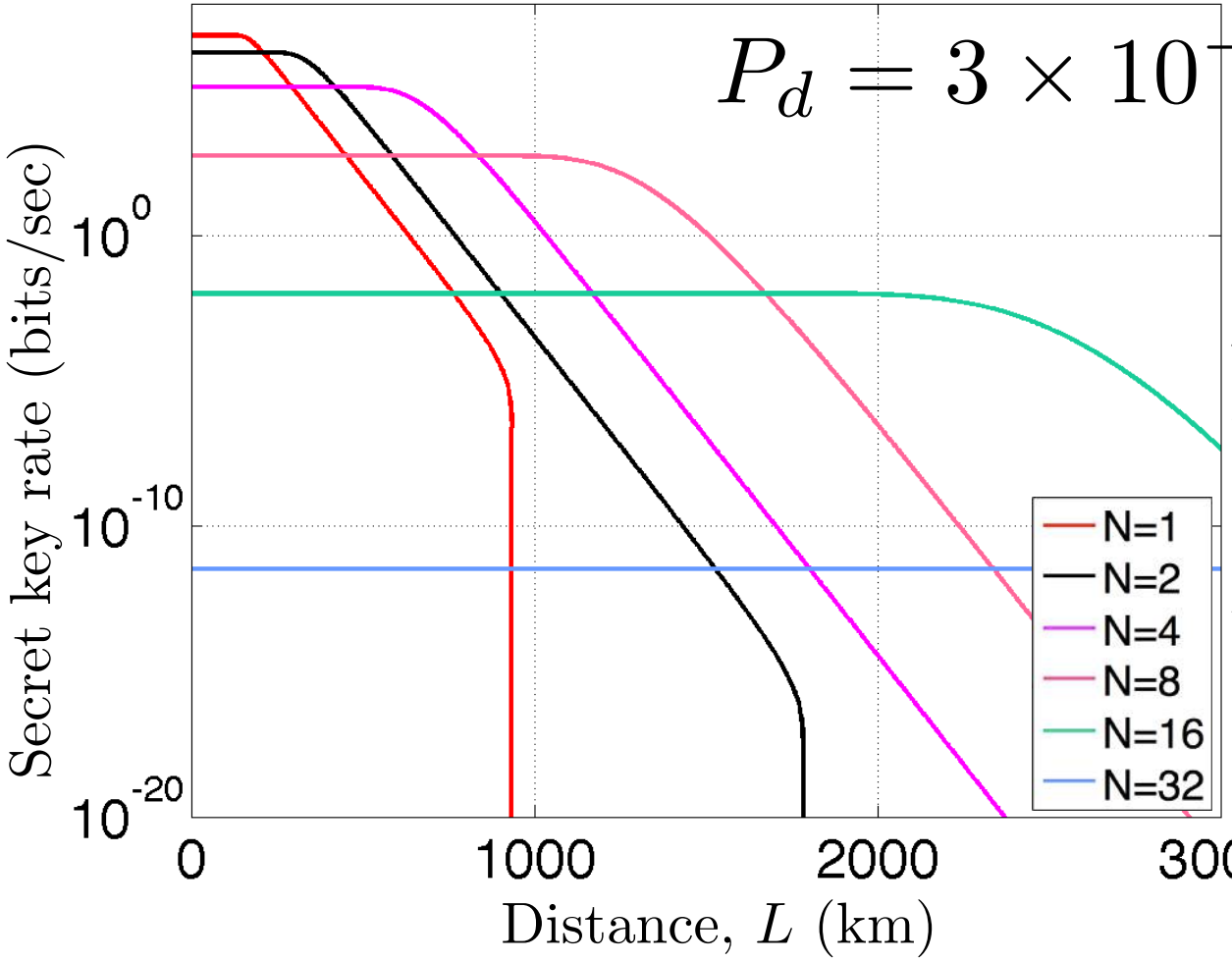
$$\begin{cases}\beta = 1 - 2P_e \\ H = 1 + t_r / \left[(1-2Q_{\text{th}})\dfrac{t_r}{t_d}\right]^{1/N}\end{cases}$$

Detector dark click rates affect the maximum range L_{max}, but largely does not affect the envelope of the secret key rates over N elementary links.

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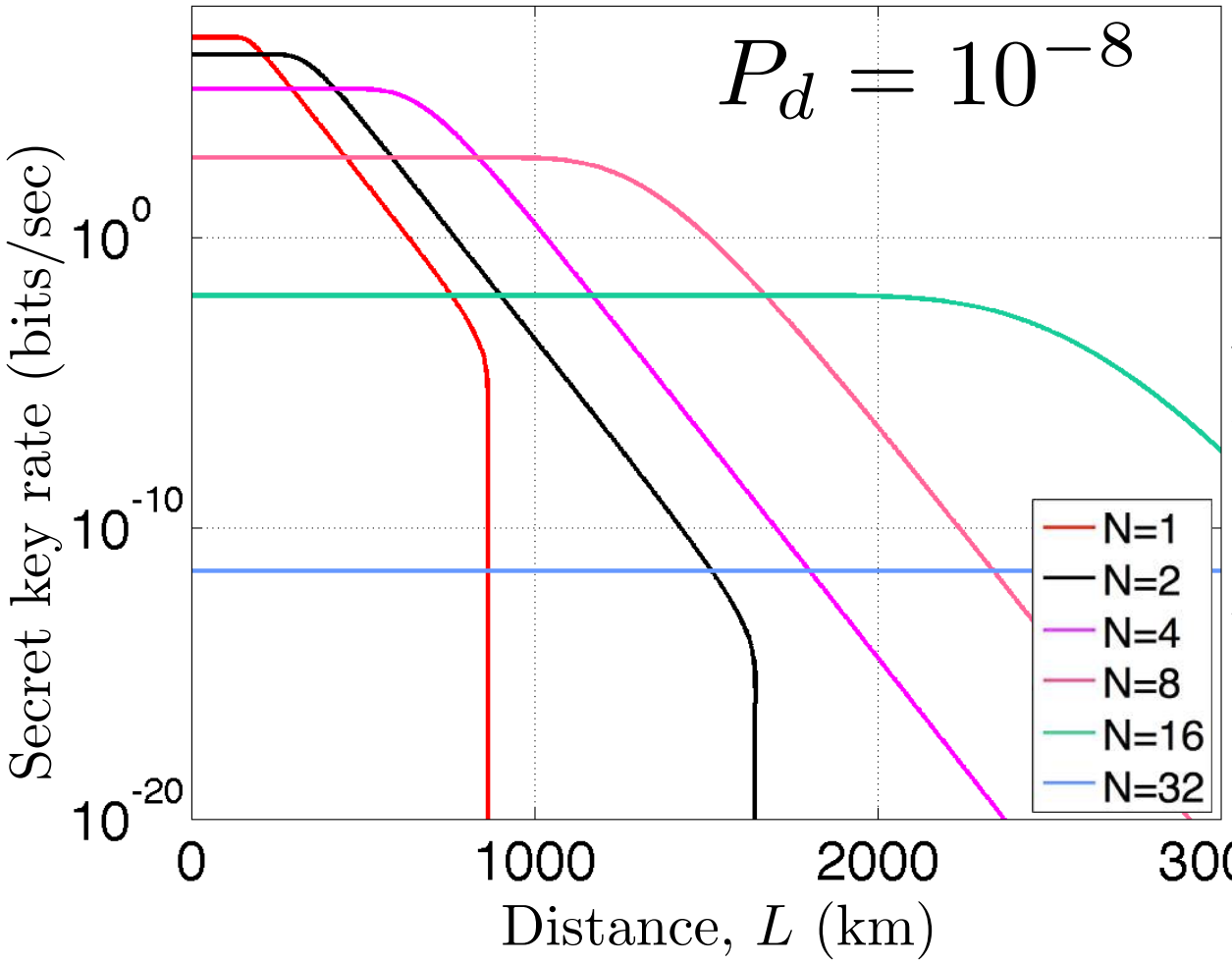
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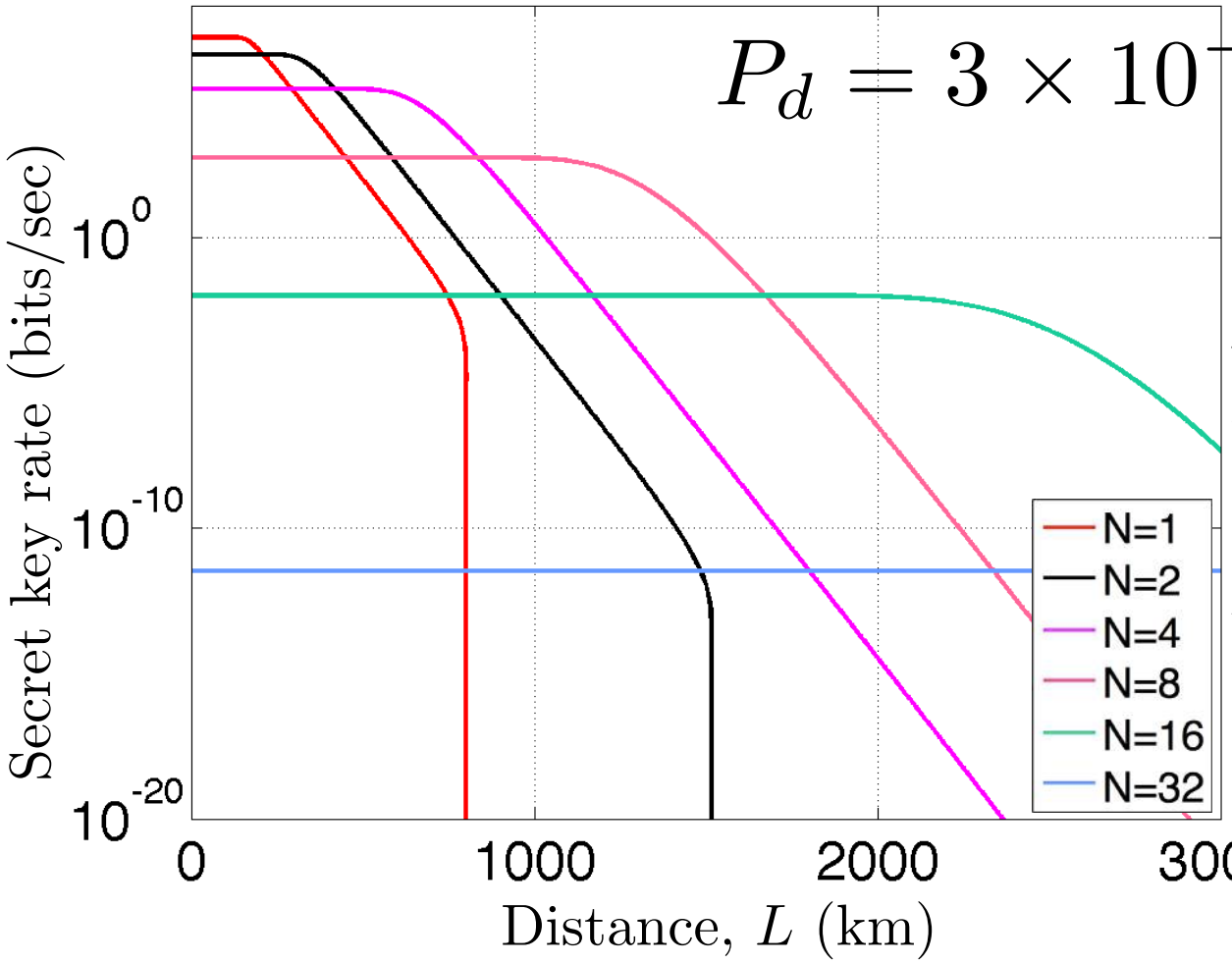
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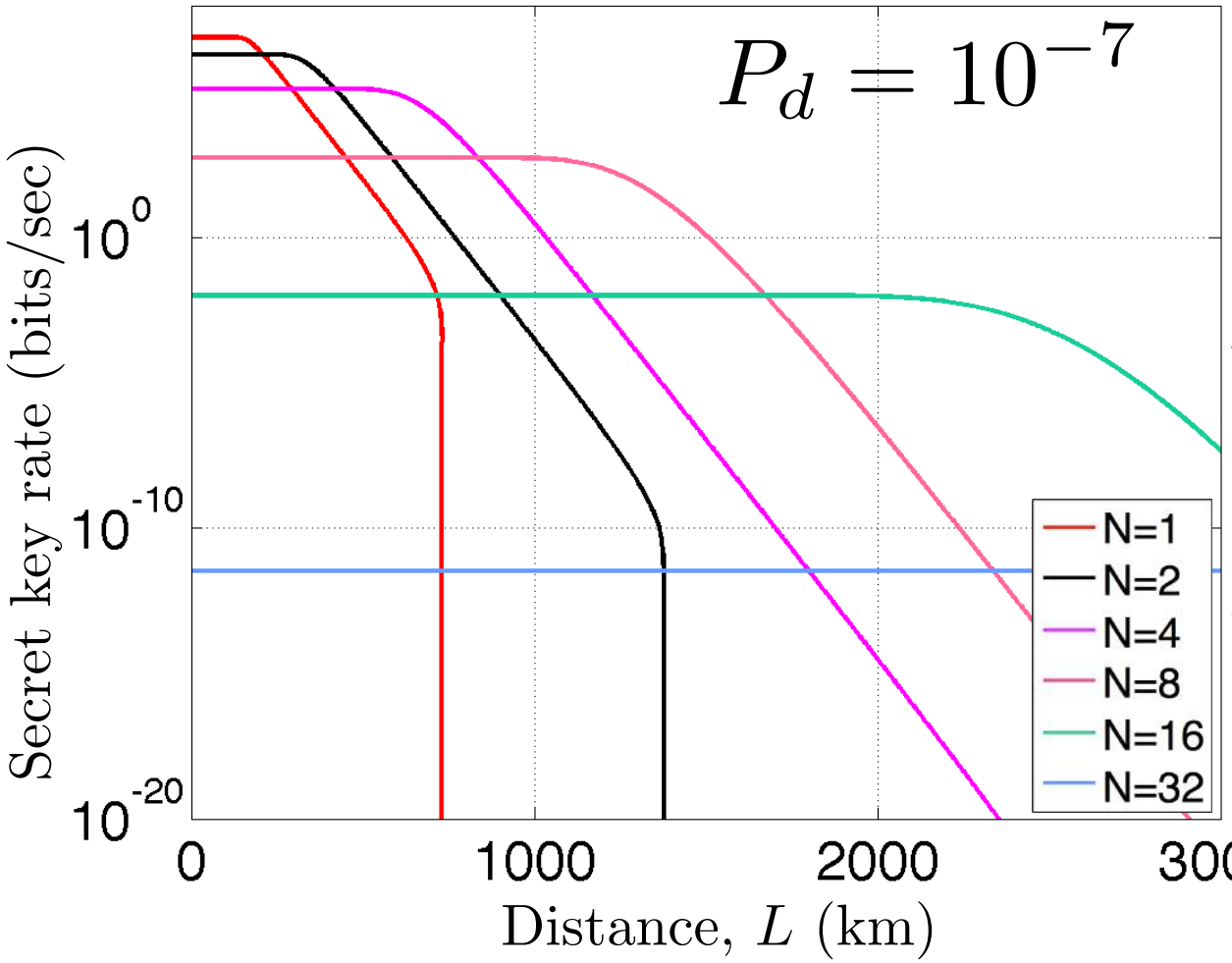
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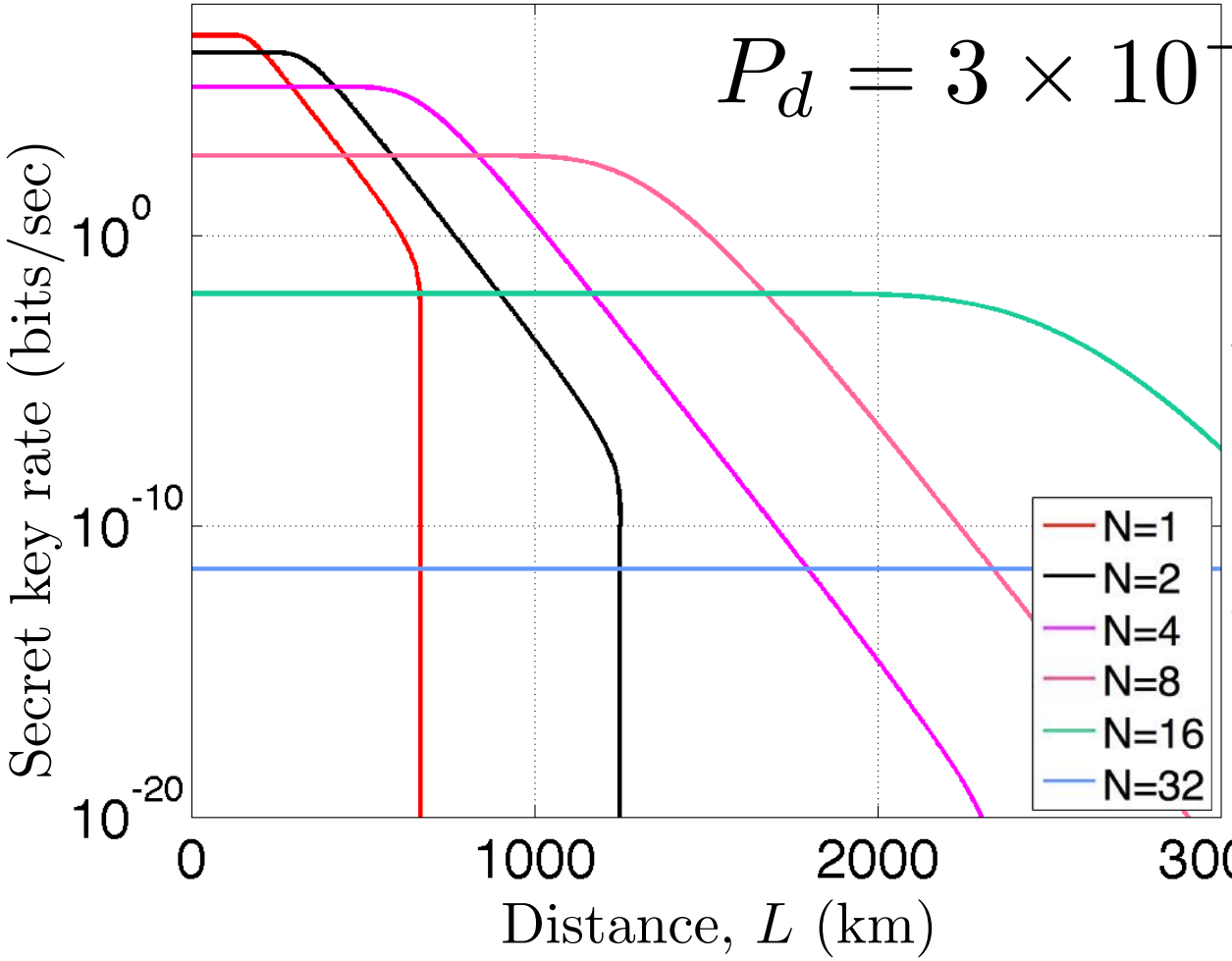
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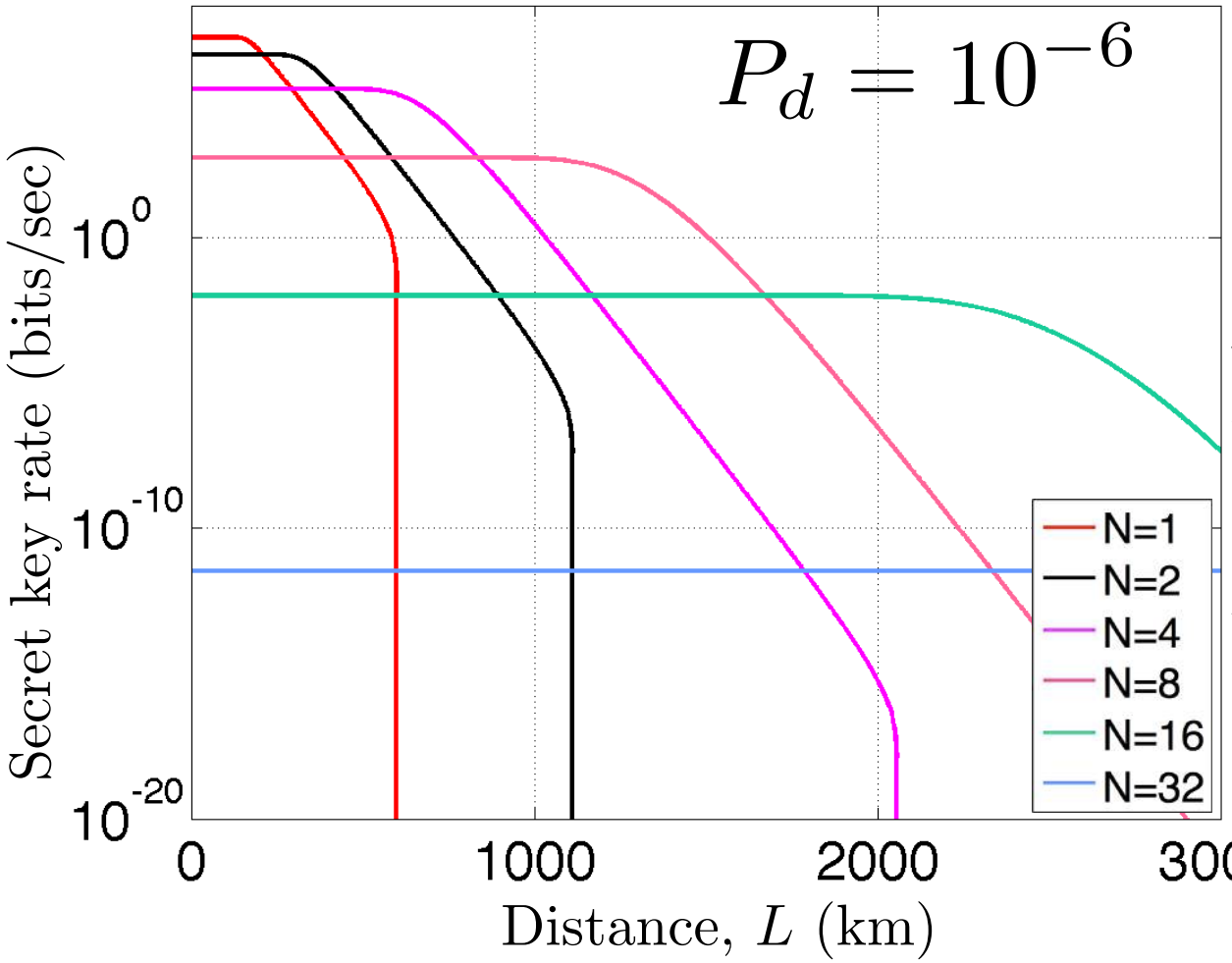
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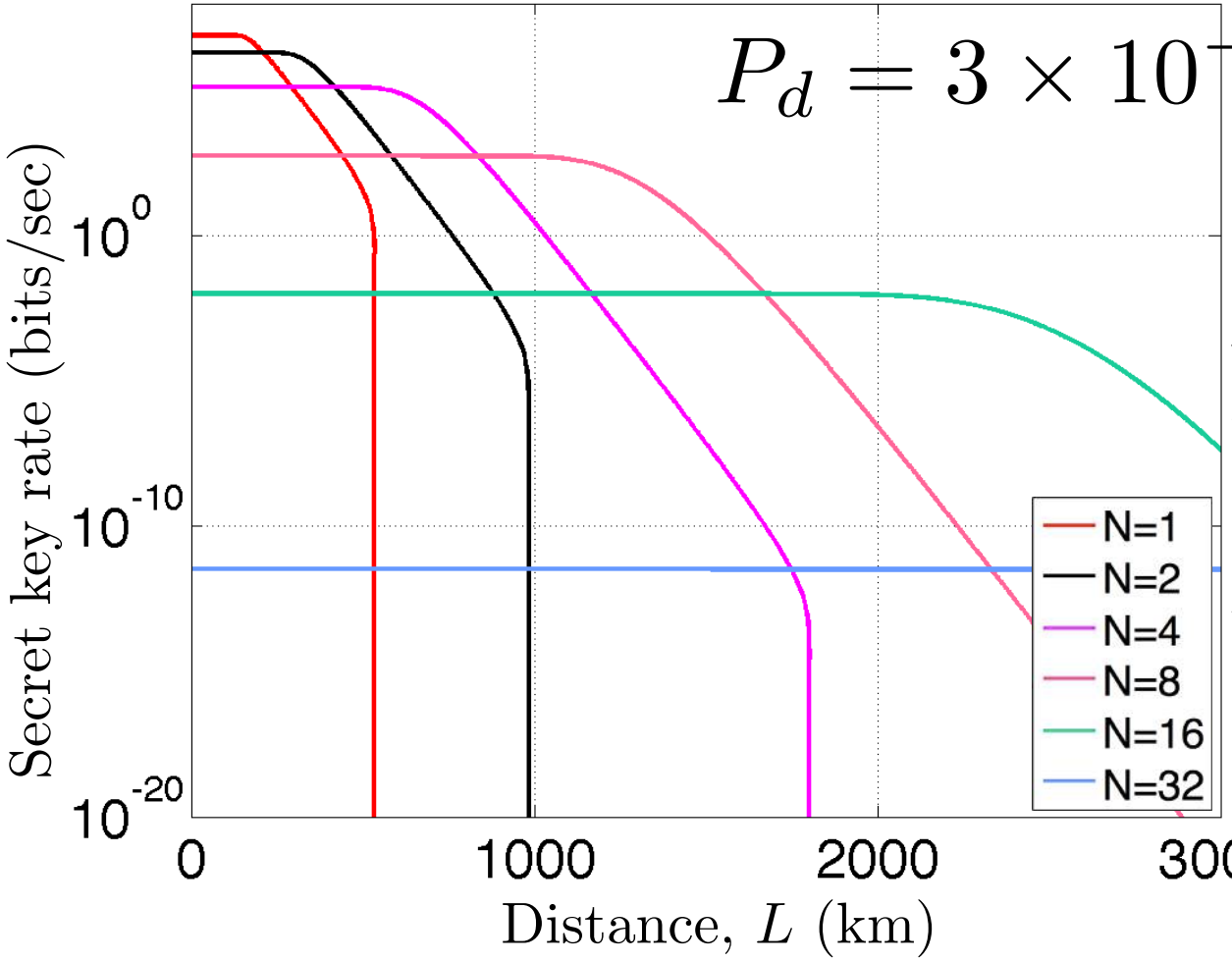
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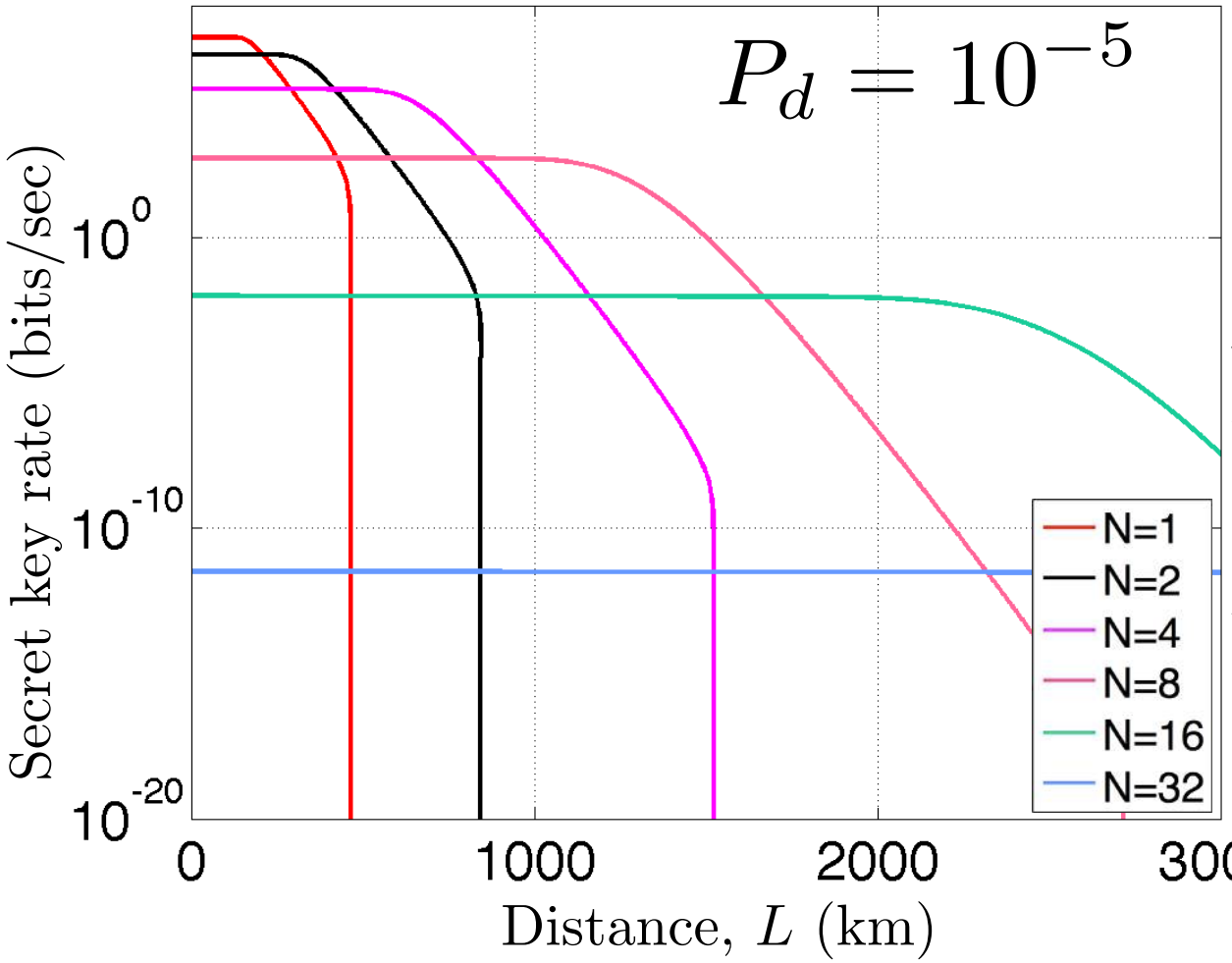
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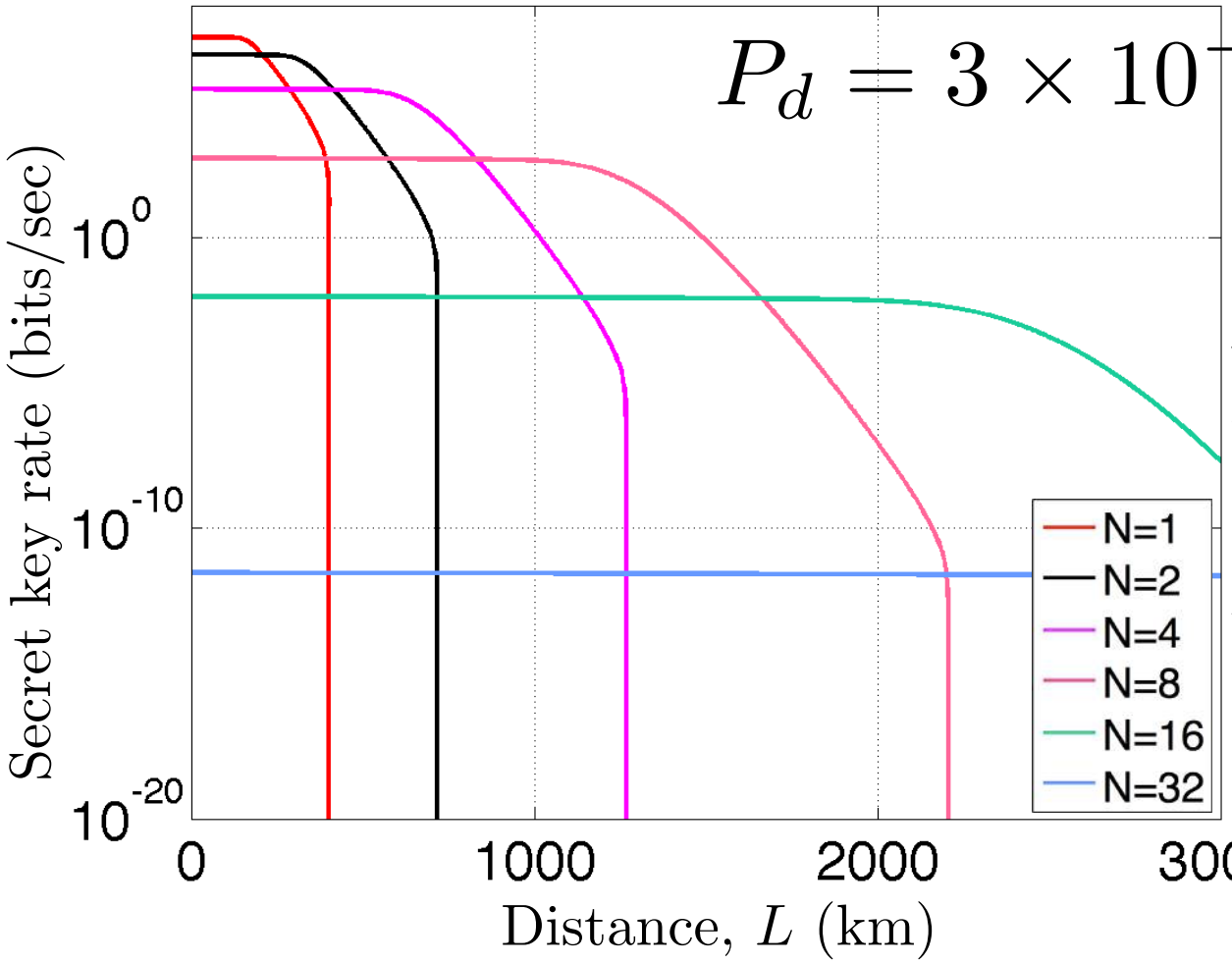
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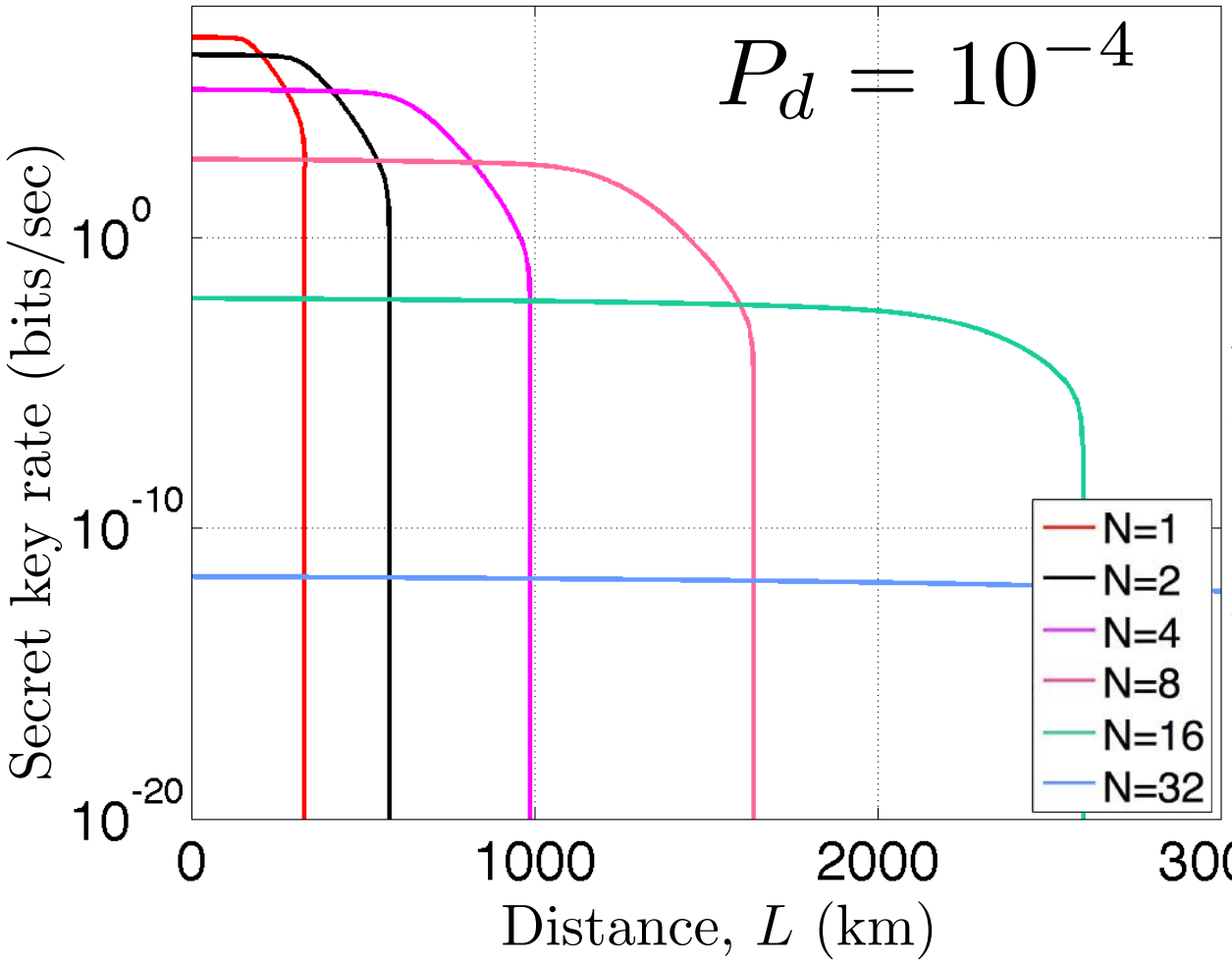
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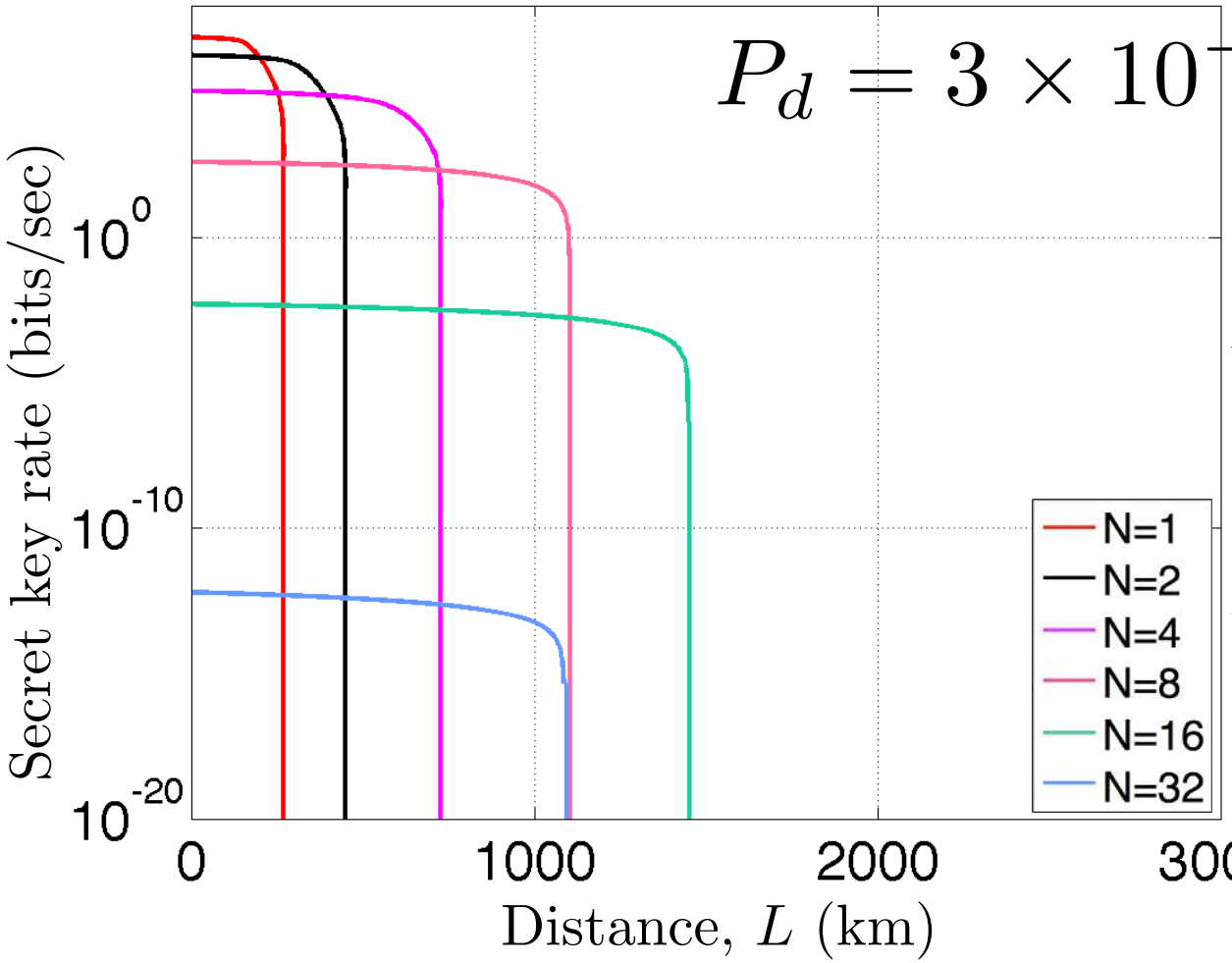
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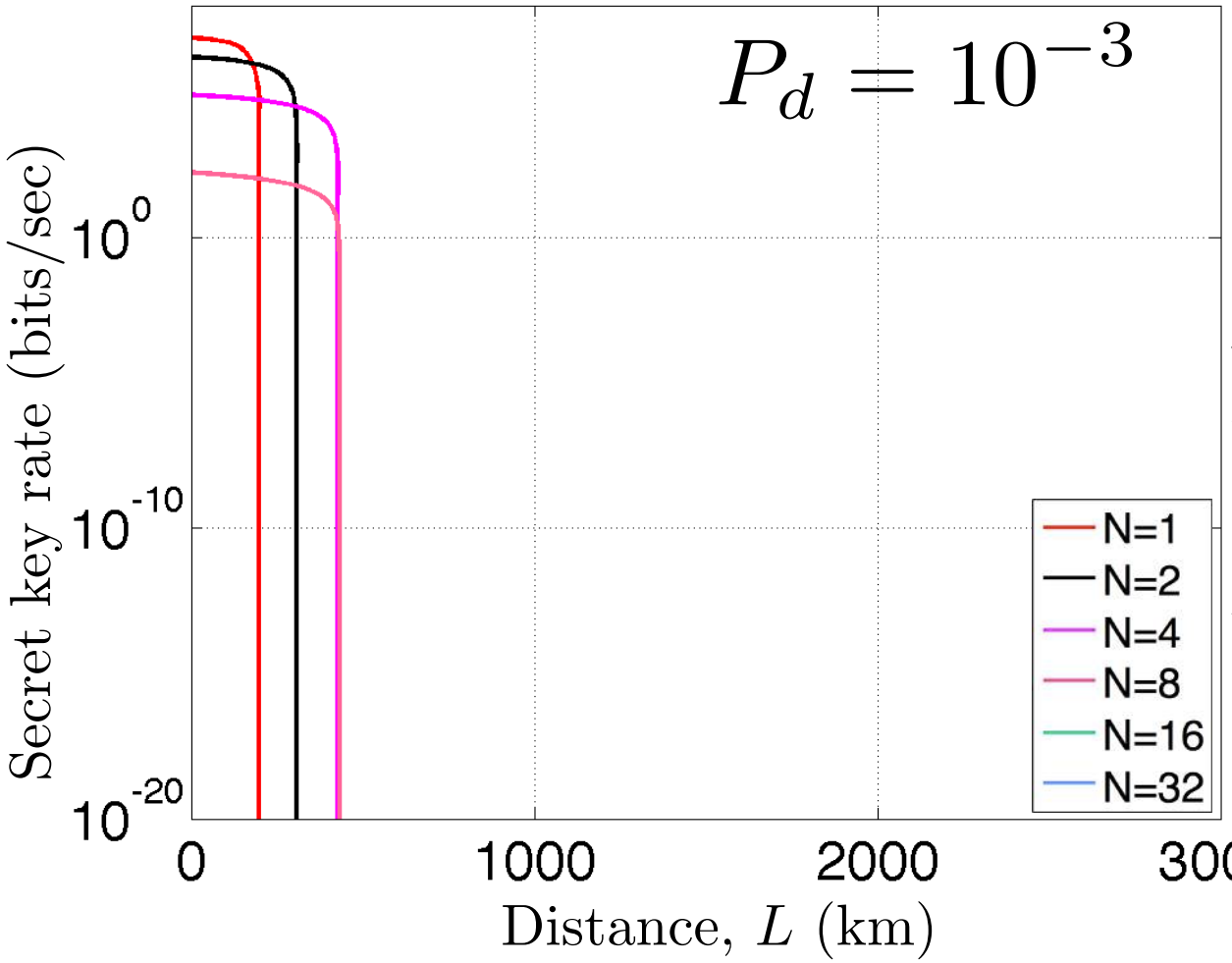
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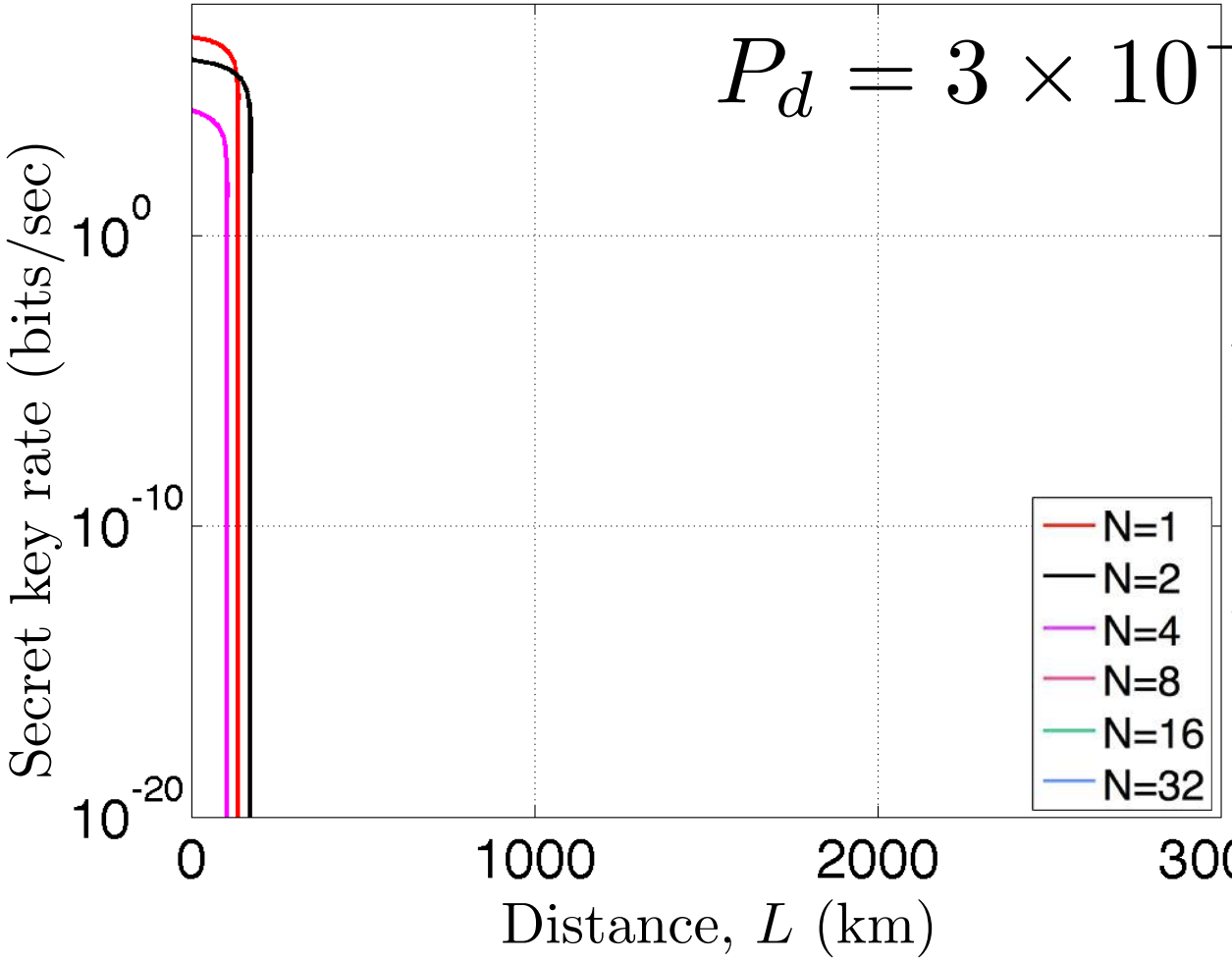
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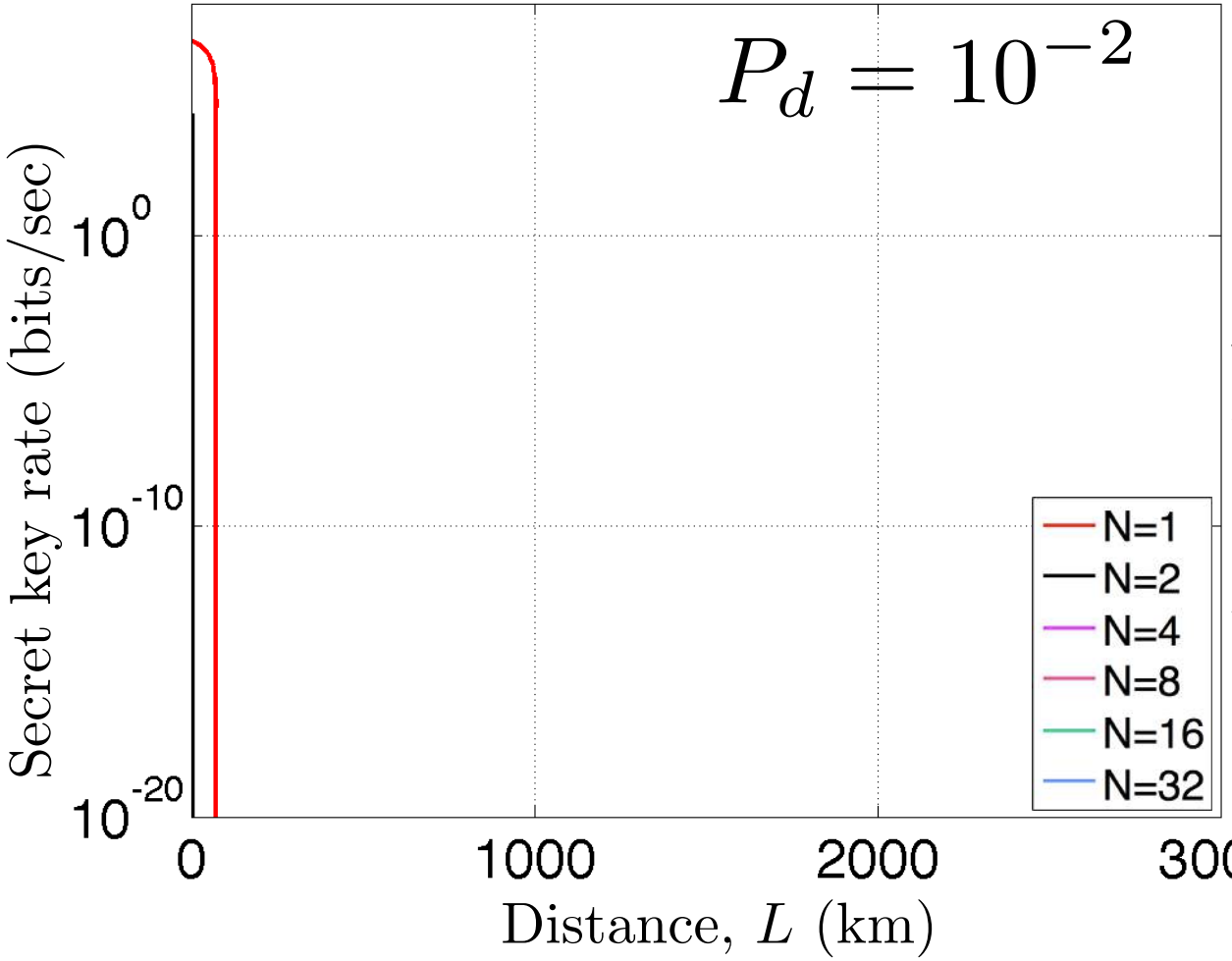
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Detector dark click rates affect the maximum range L_{max}, but largely does not affect the envelope of the secret key rates over N elementary links.

Effect of dark-click probability on the rate-distance performance

Raytheon BBN Technologies

- Assume source has zero two-photon emissions



$$\eta_r = \eta_d = \eta_e = 0.9$$

$$T_q = 50\,\text{ns}$$

$$\alpha = 0.15\,\text{dB/km}$$

$$\lambda_m = 10^{-0.1\gamma_m}, \gamma = 1\,\text{dB}$$

$$M = 1000\,\text{frequency modes}$$

$$P_d = P_r = P_e$$

$N =$ number of elementary links

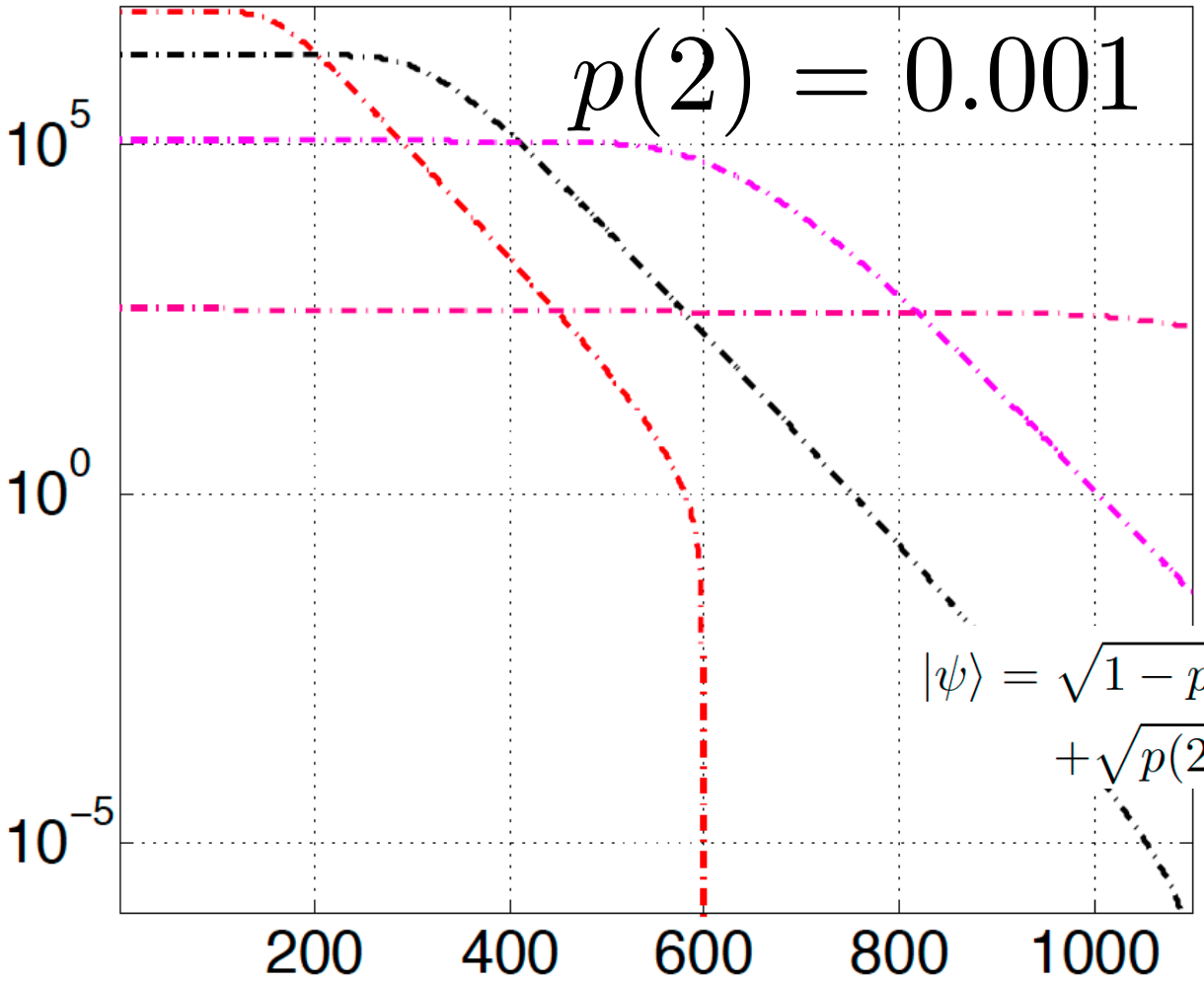
$$L_{\max} = \frac{20N}{\alpha}\log_{10}\left[\frac{\eta_e}{8P_e}\left(\sqrt{8\beta H} - 4\beta\right)\right]$$

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Effect of multi-photon emission on the rate-distance performance

Raytheon
BBN Technologies



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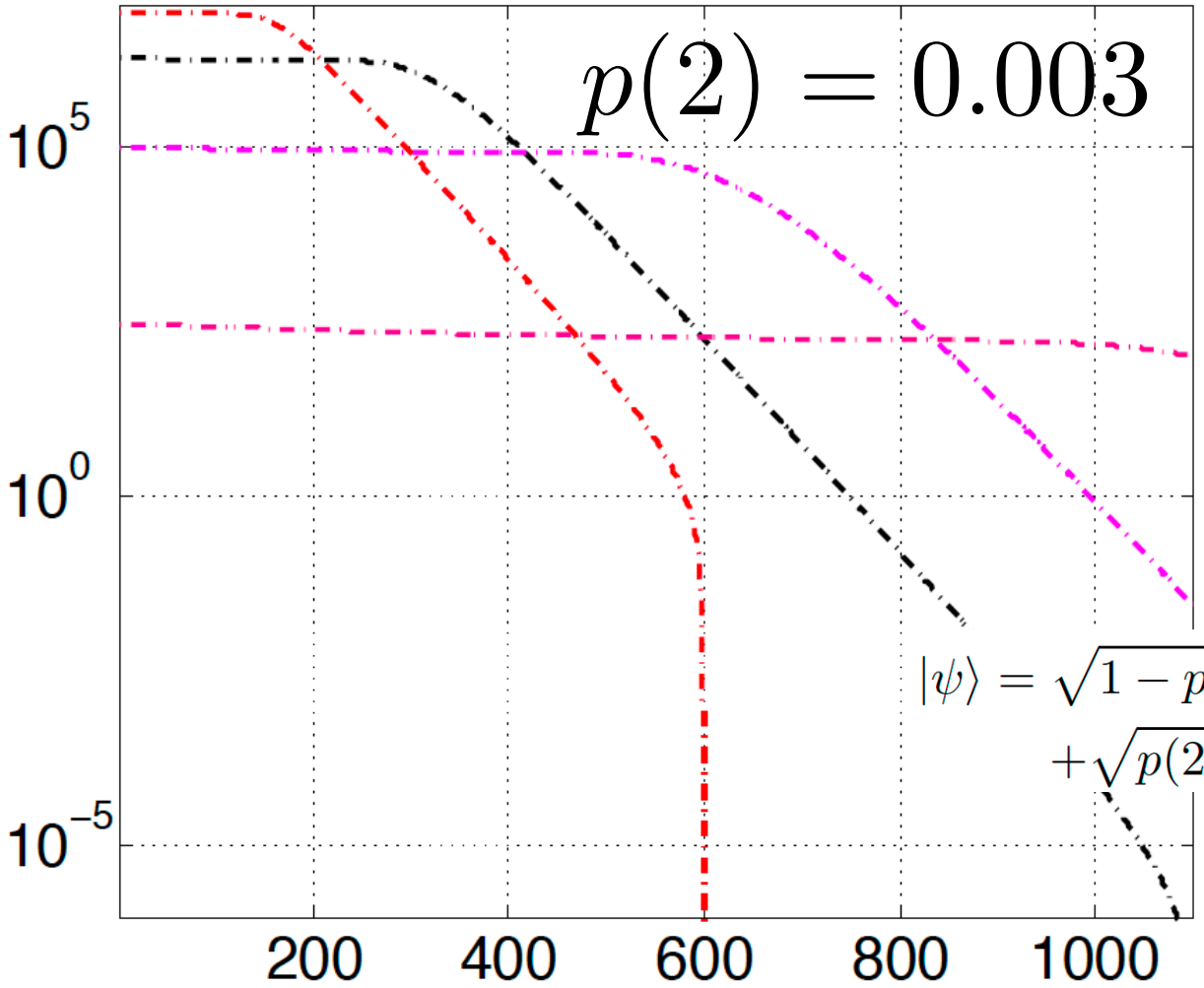
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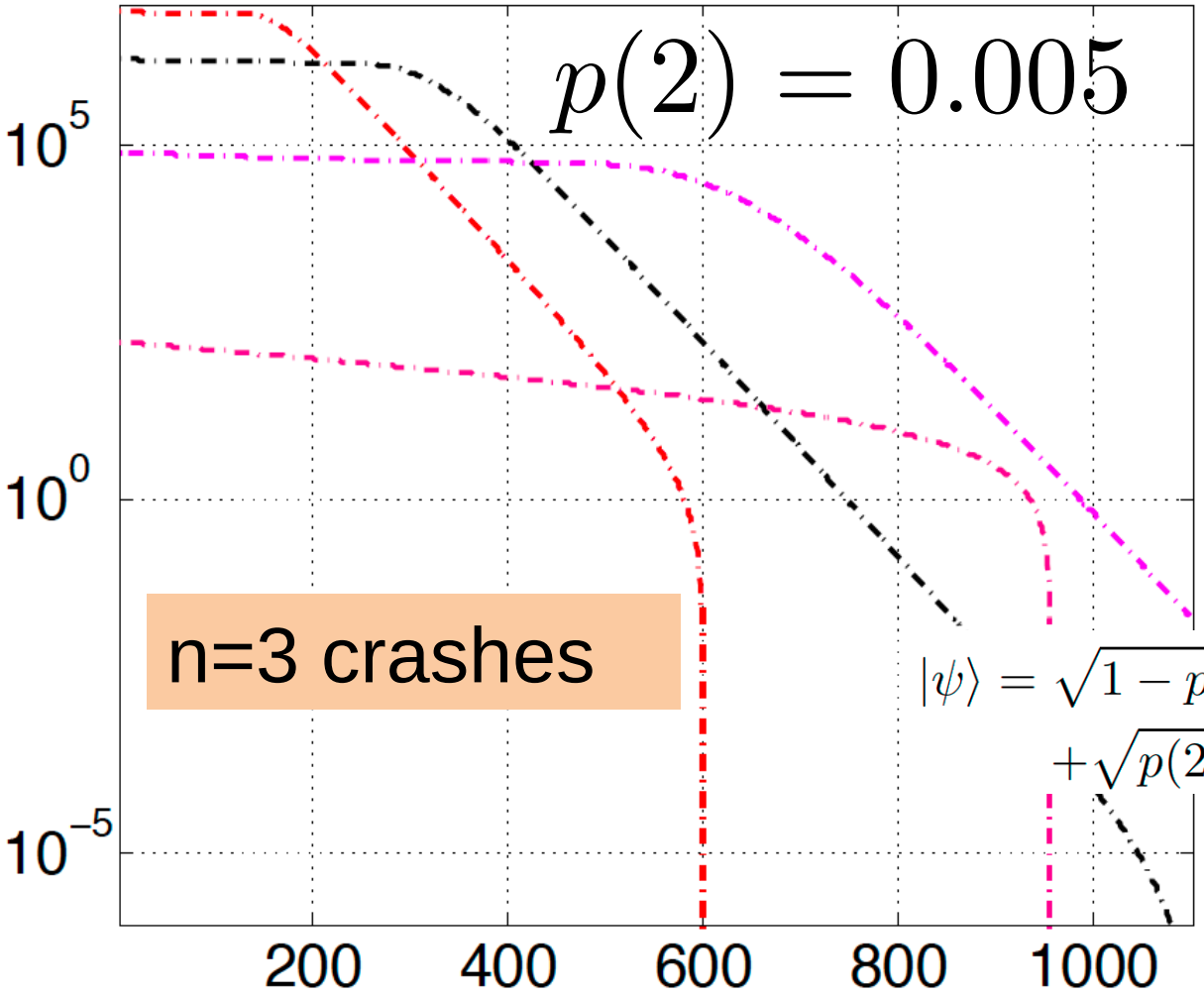
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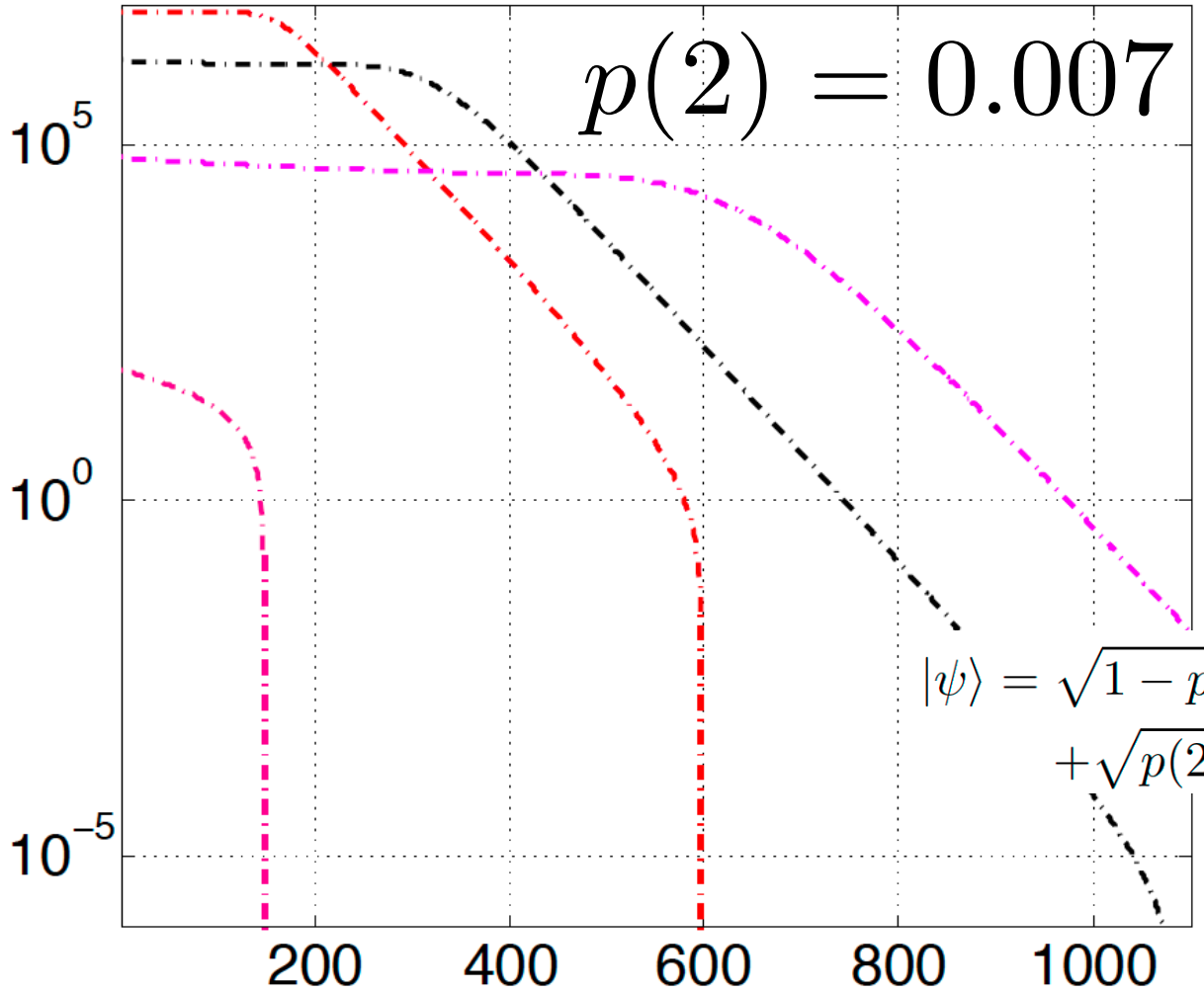
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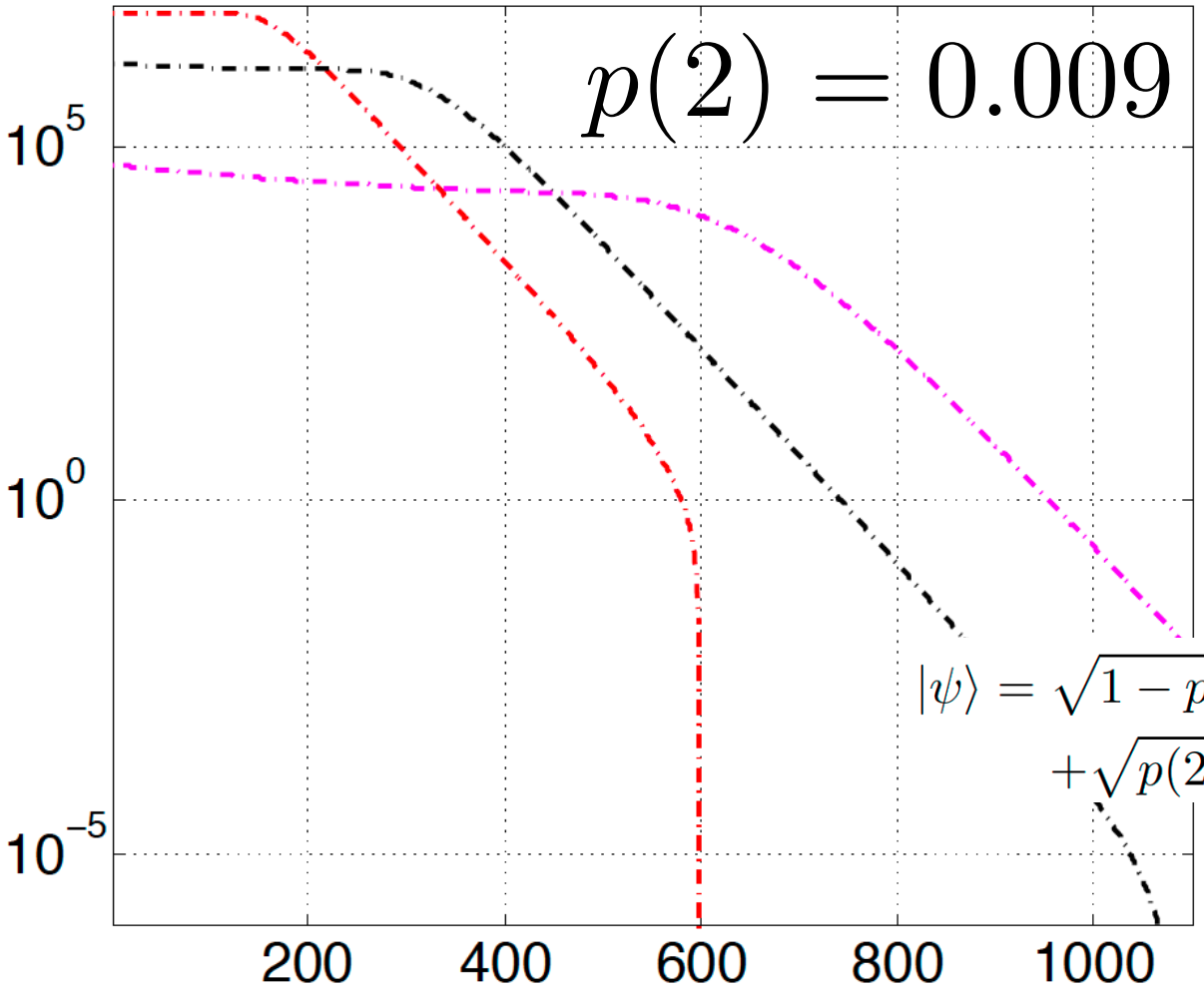
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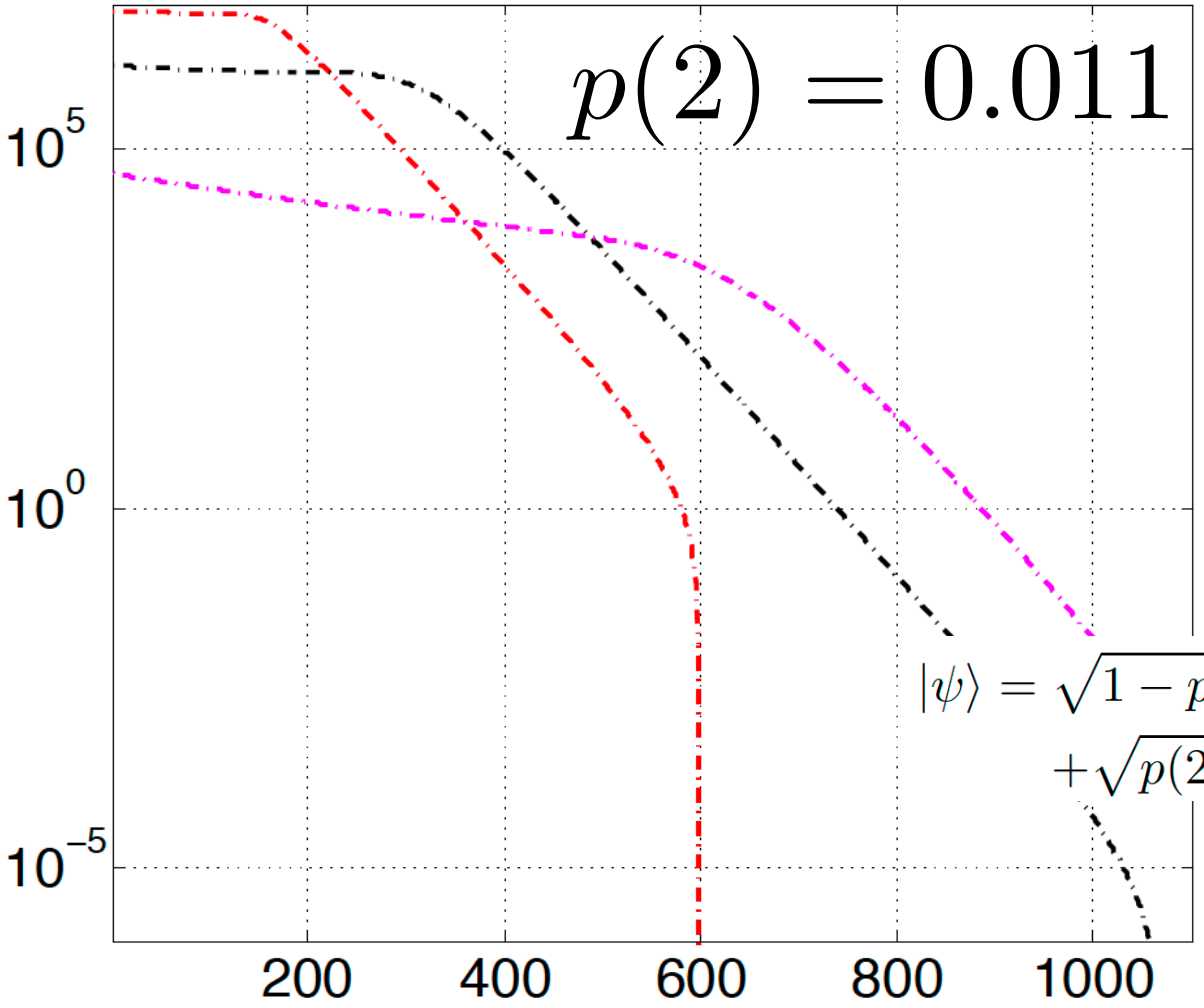
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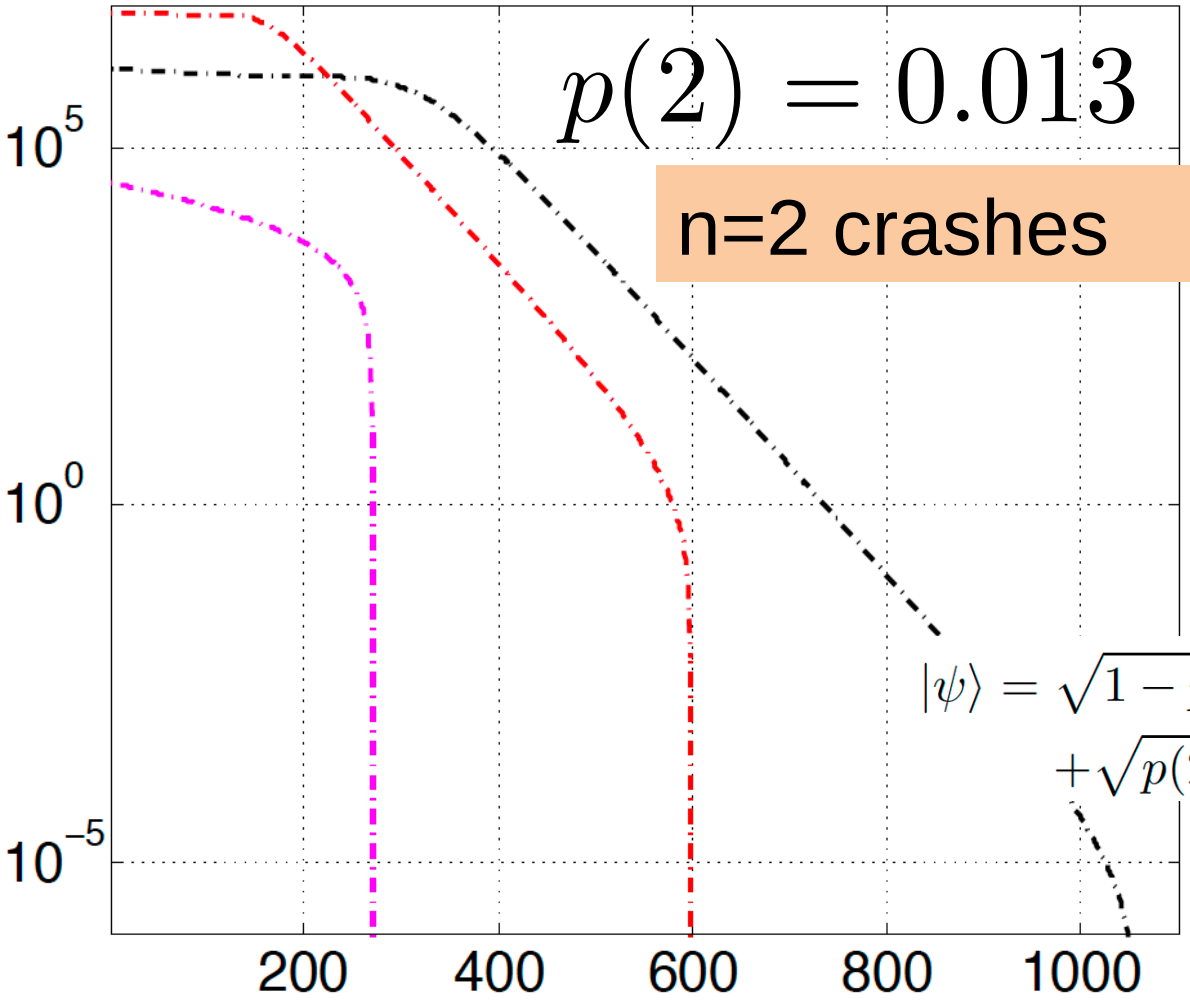
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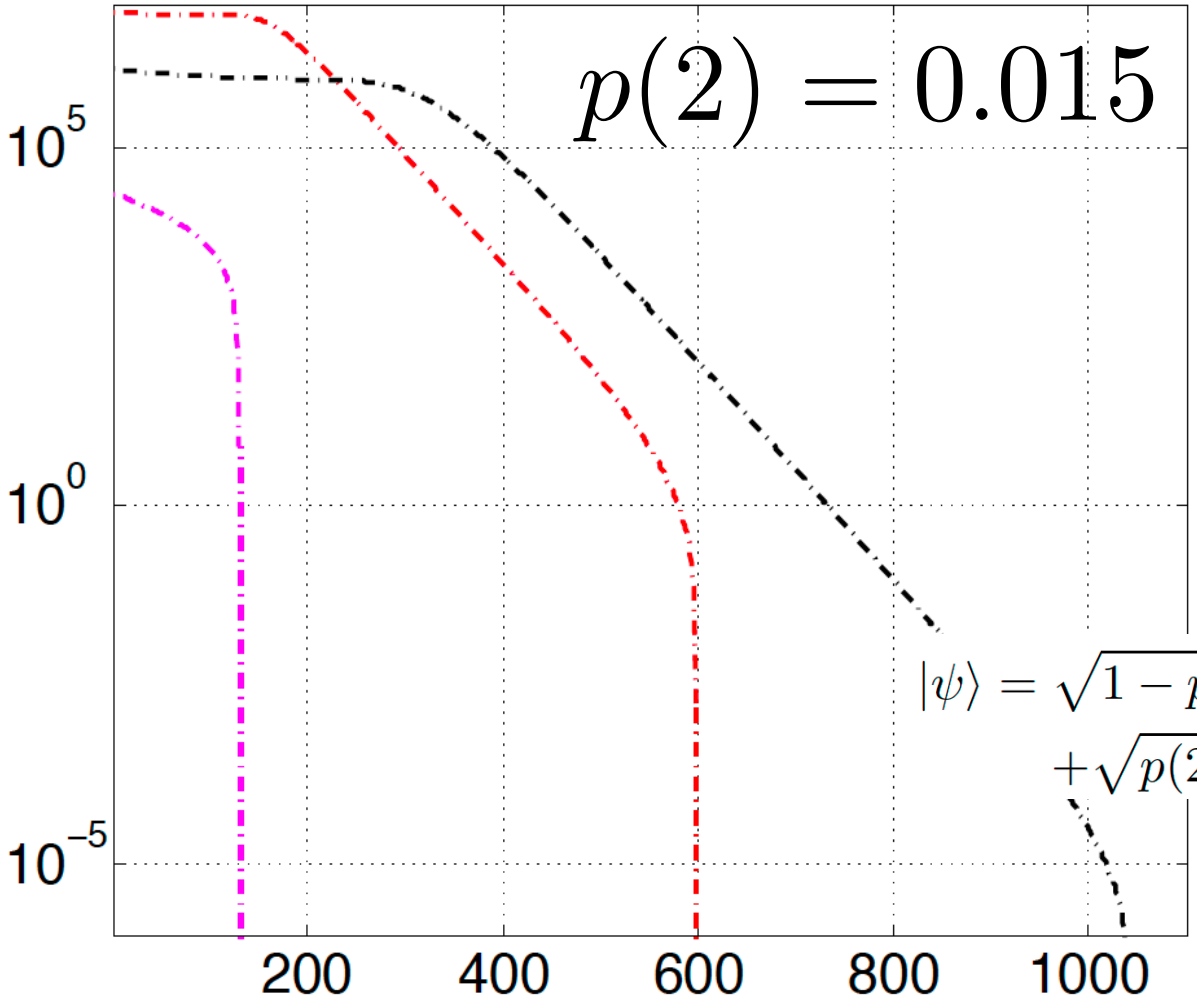
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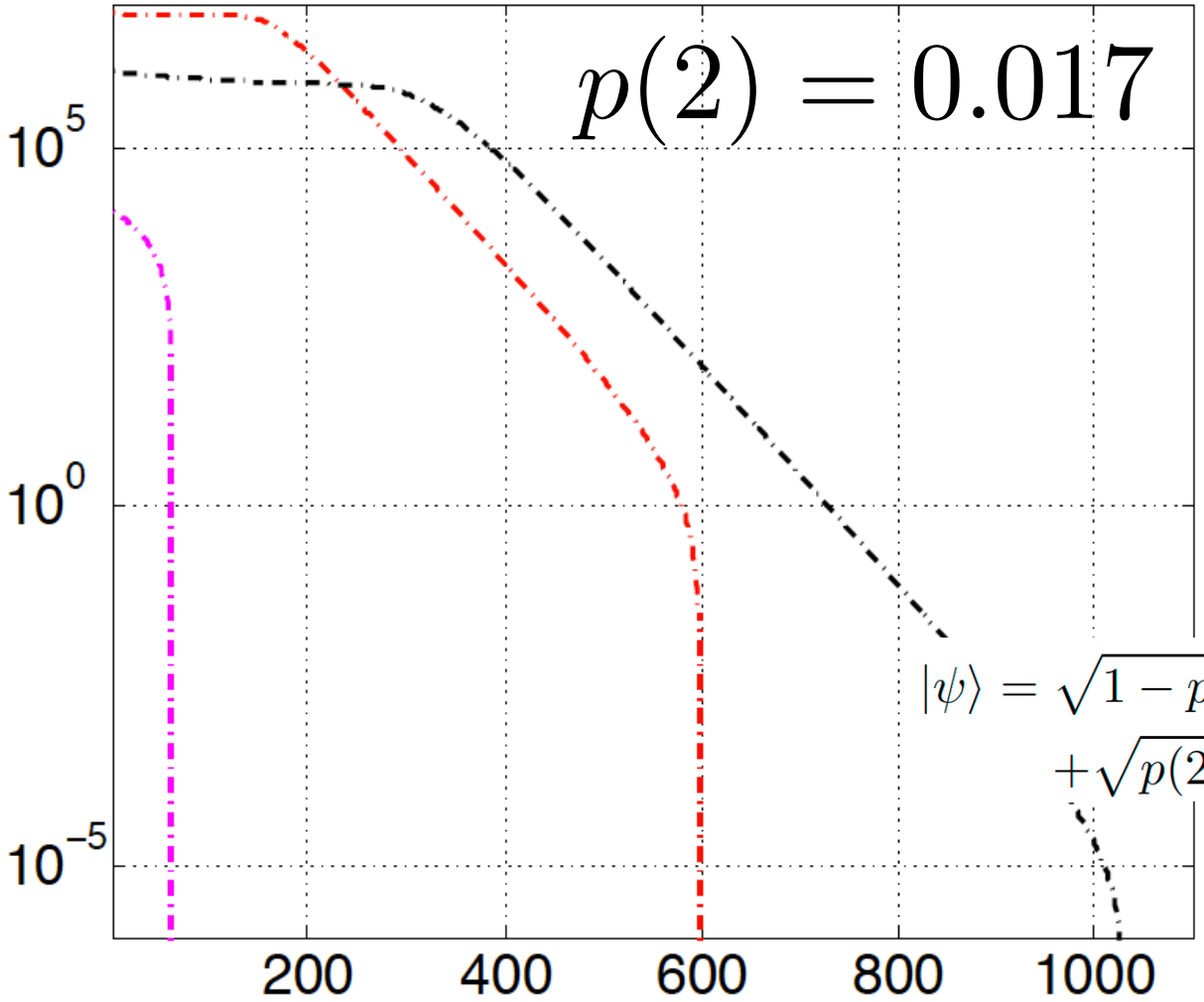
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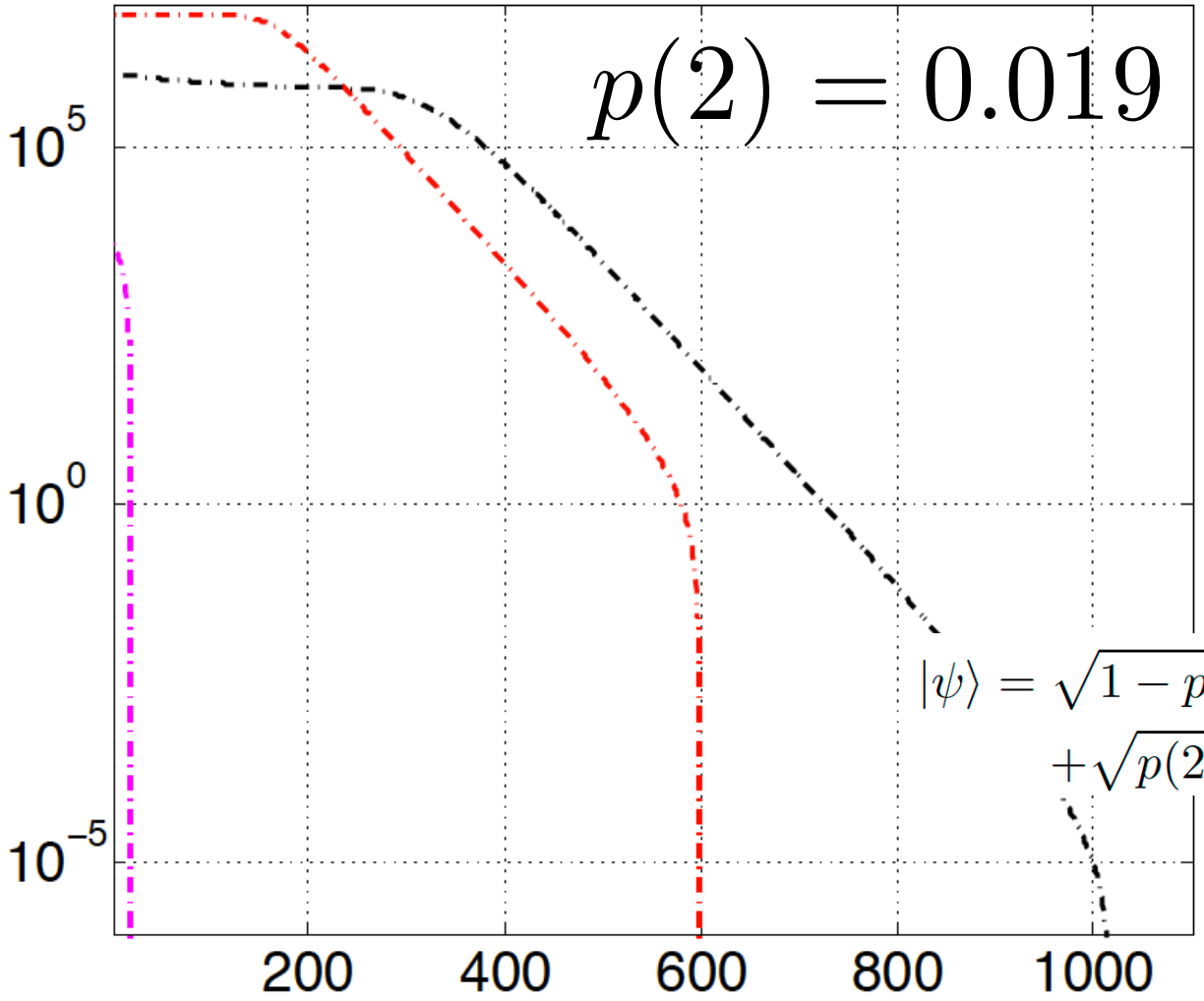
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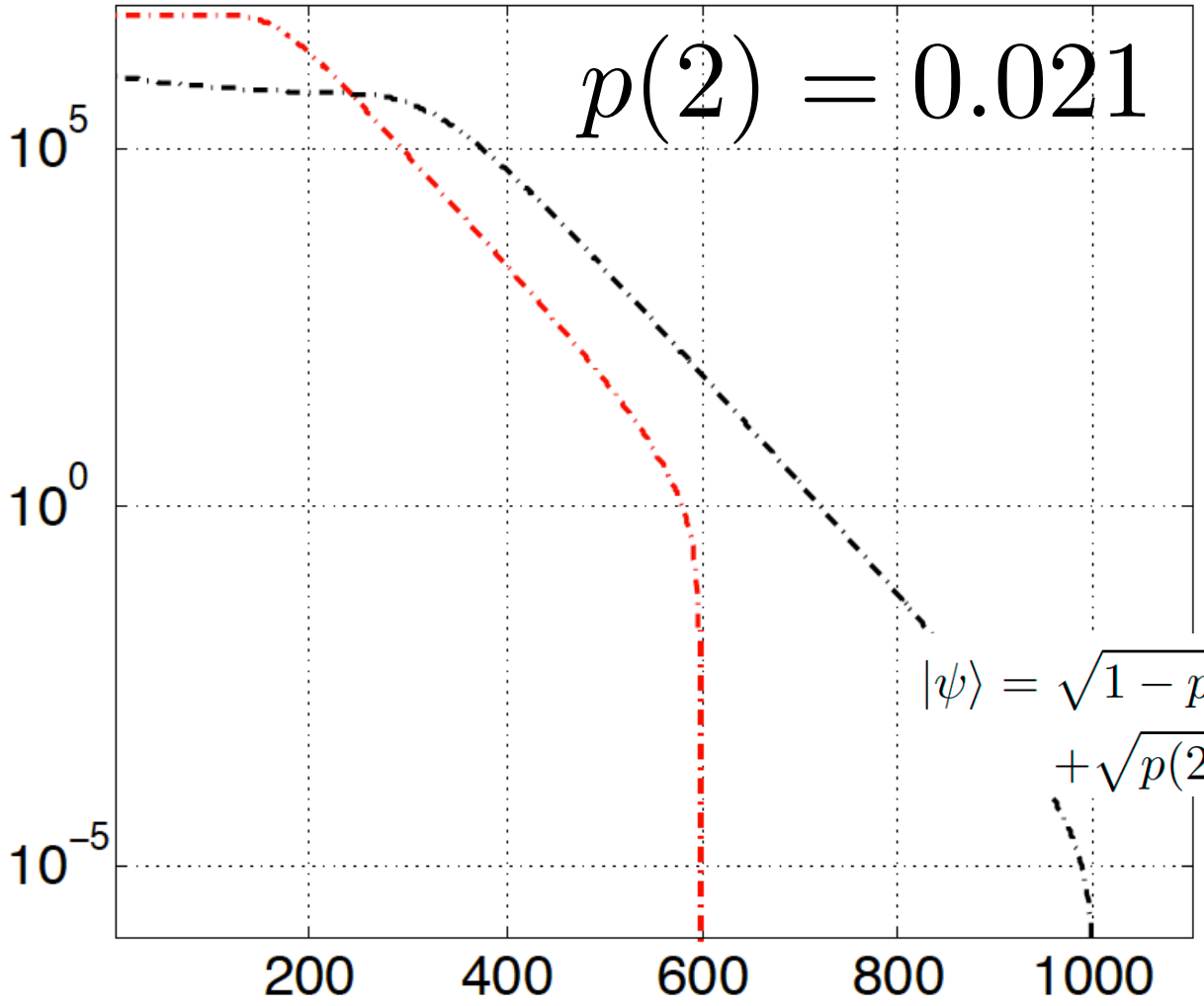
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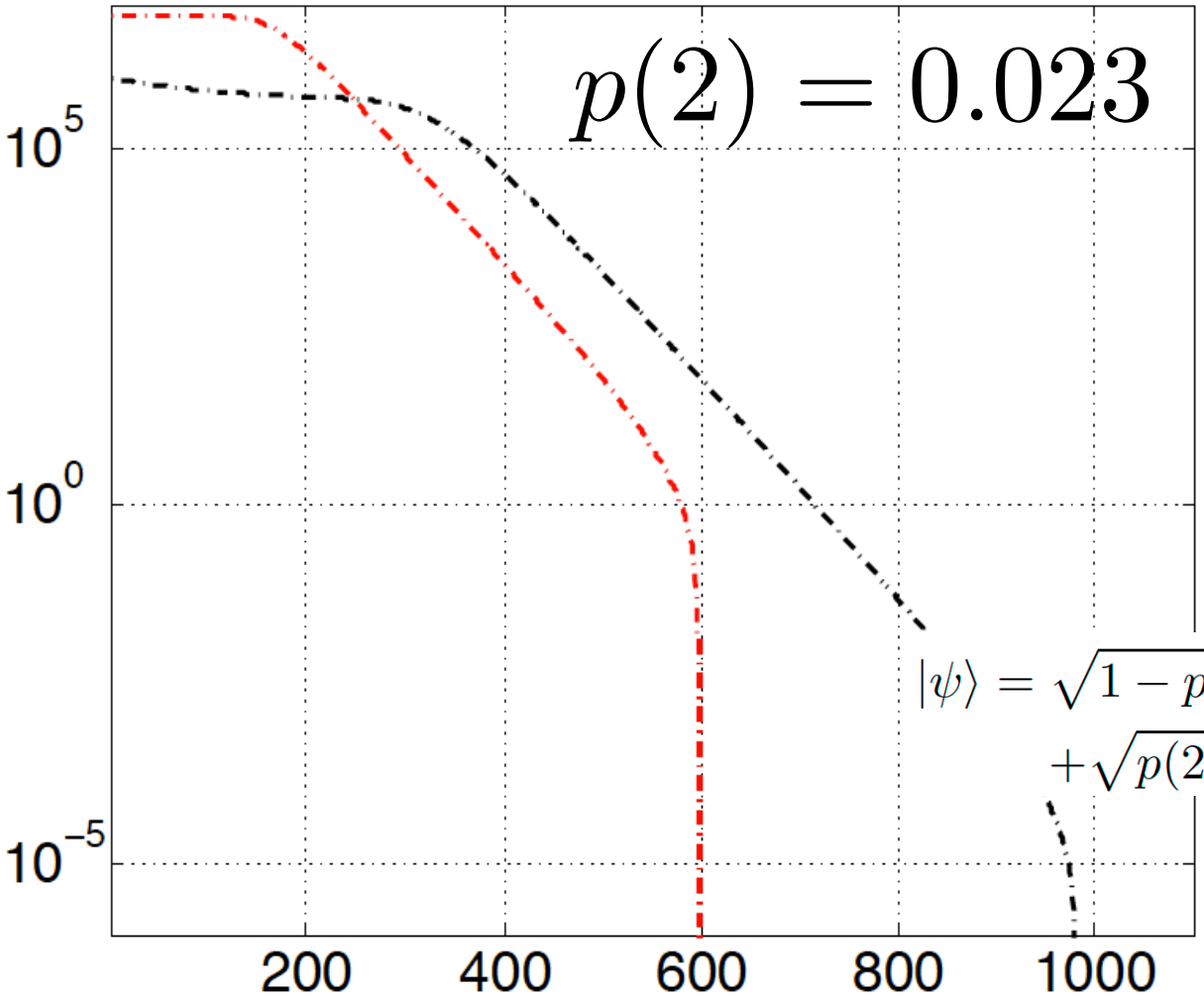
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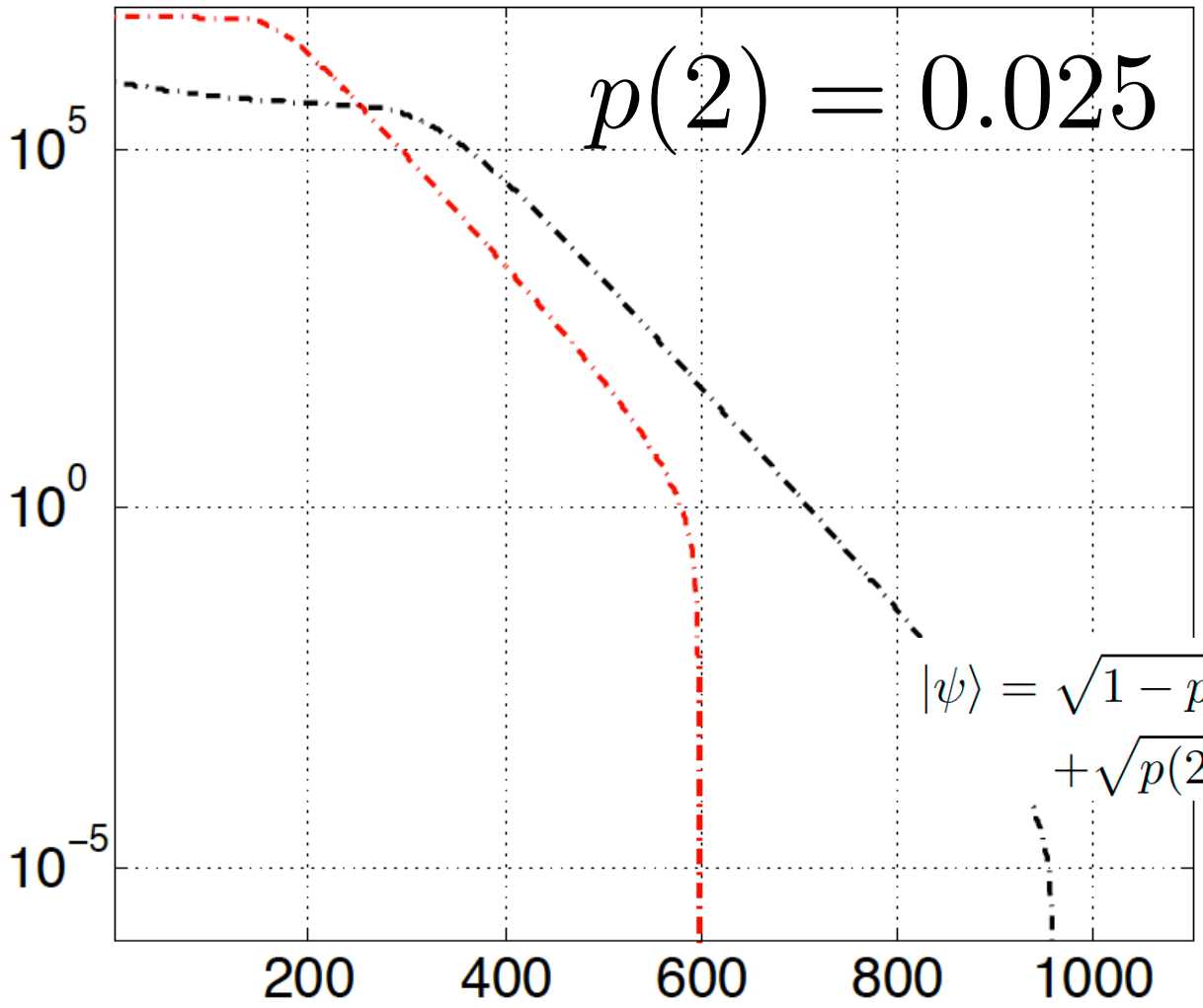
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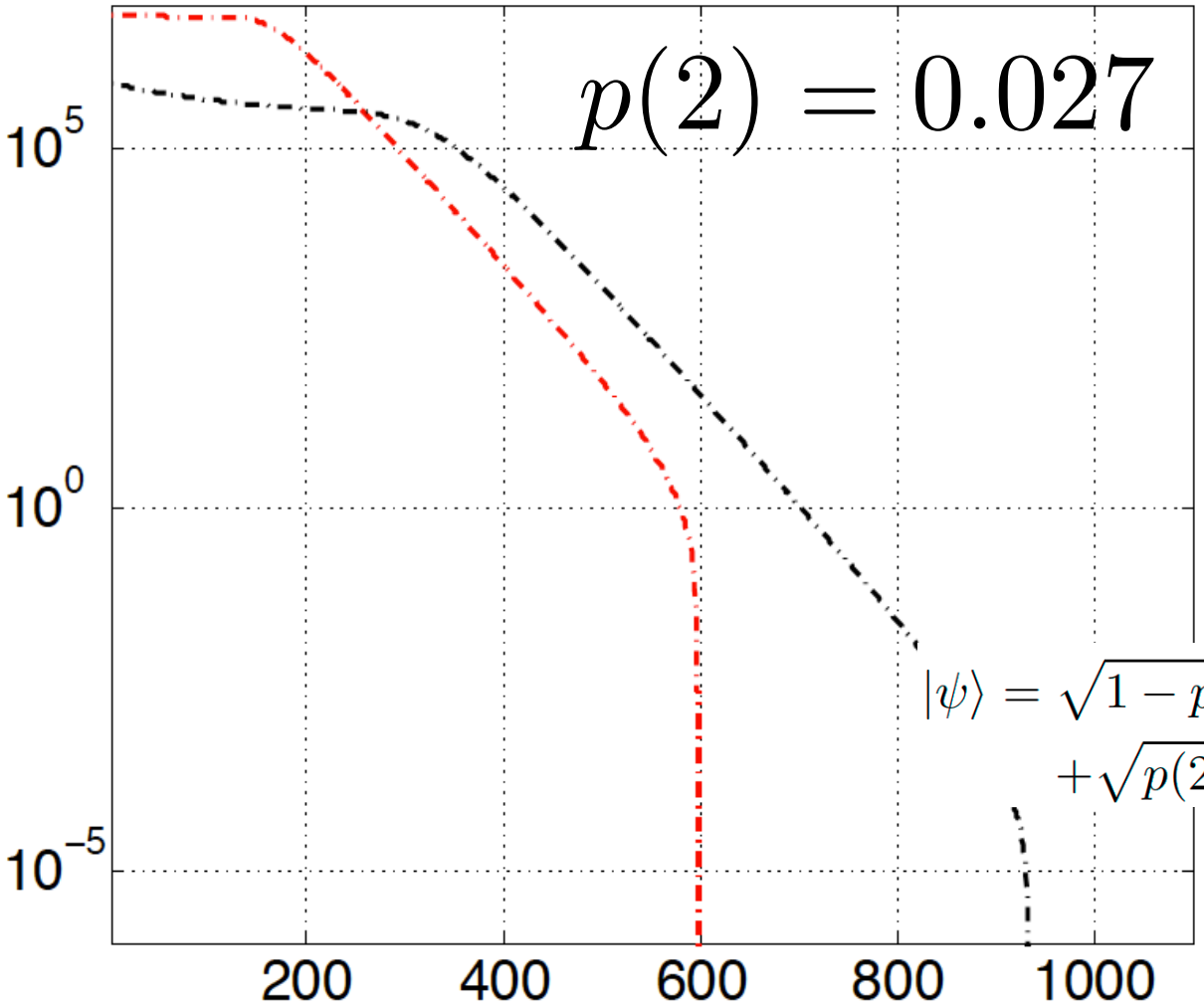
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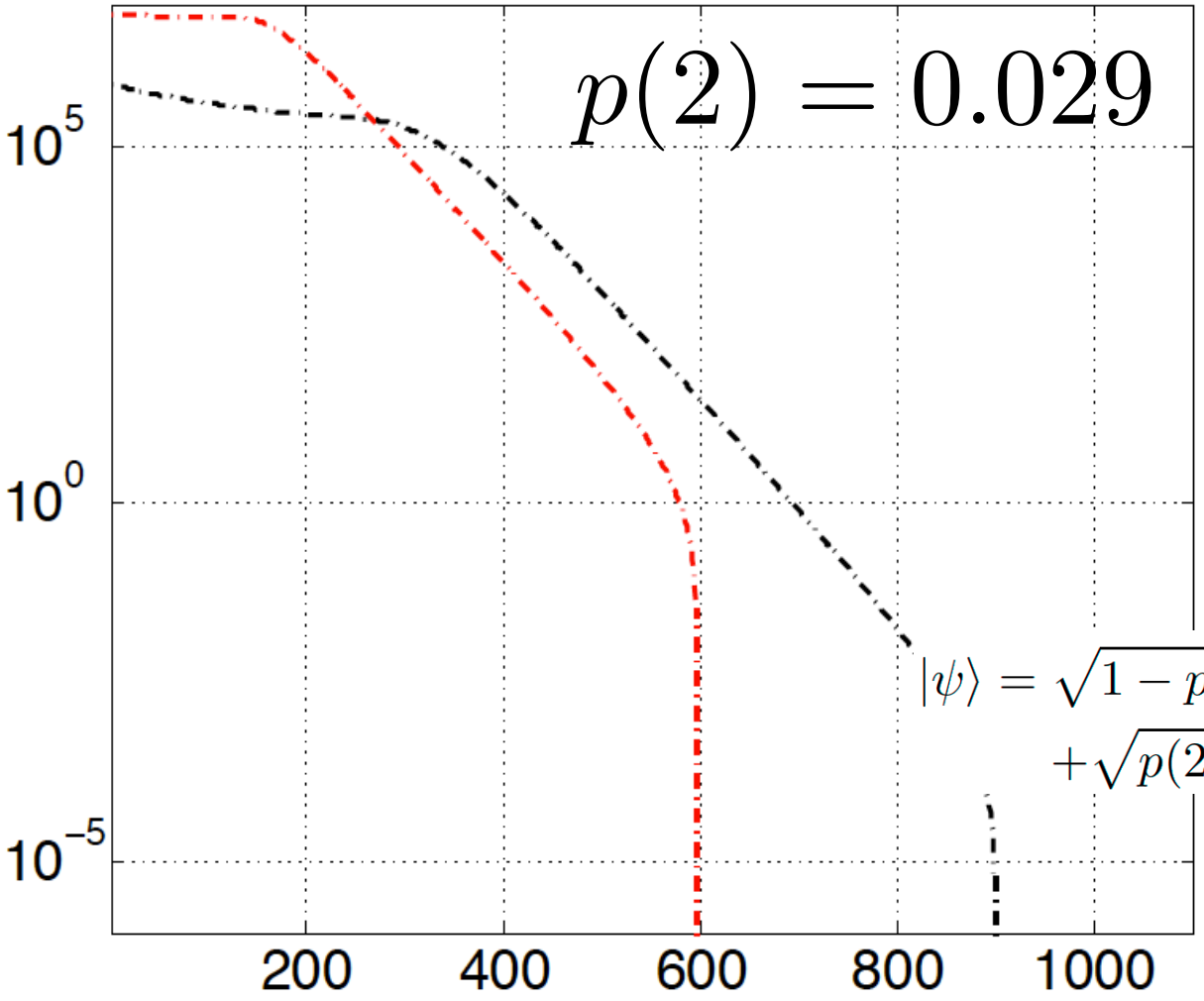
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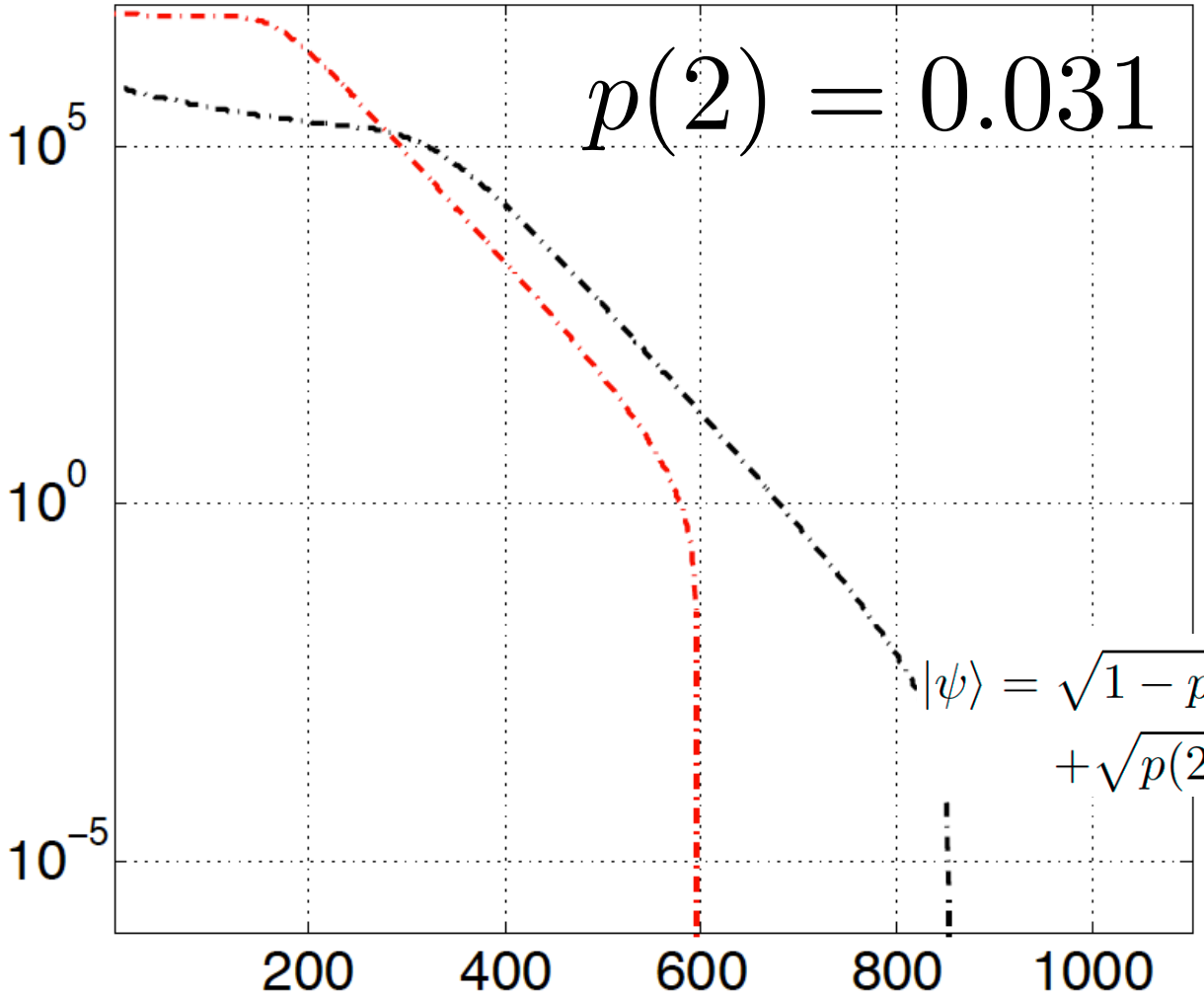
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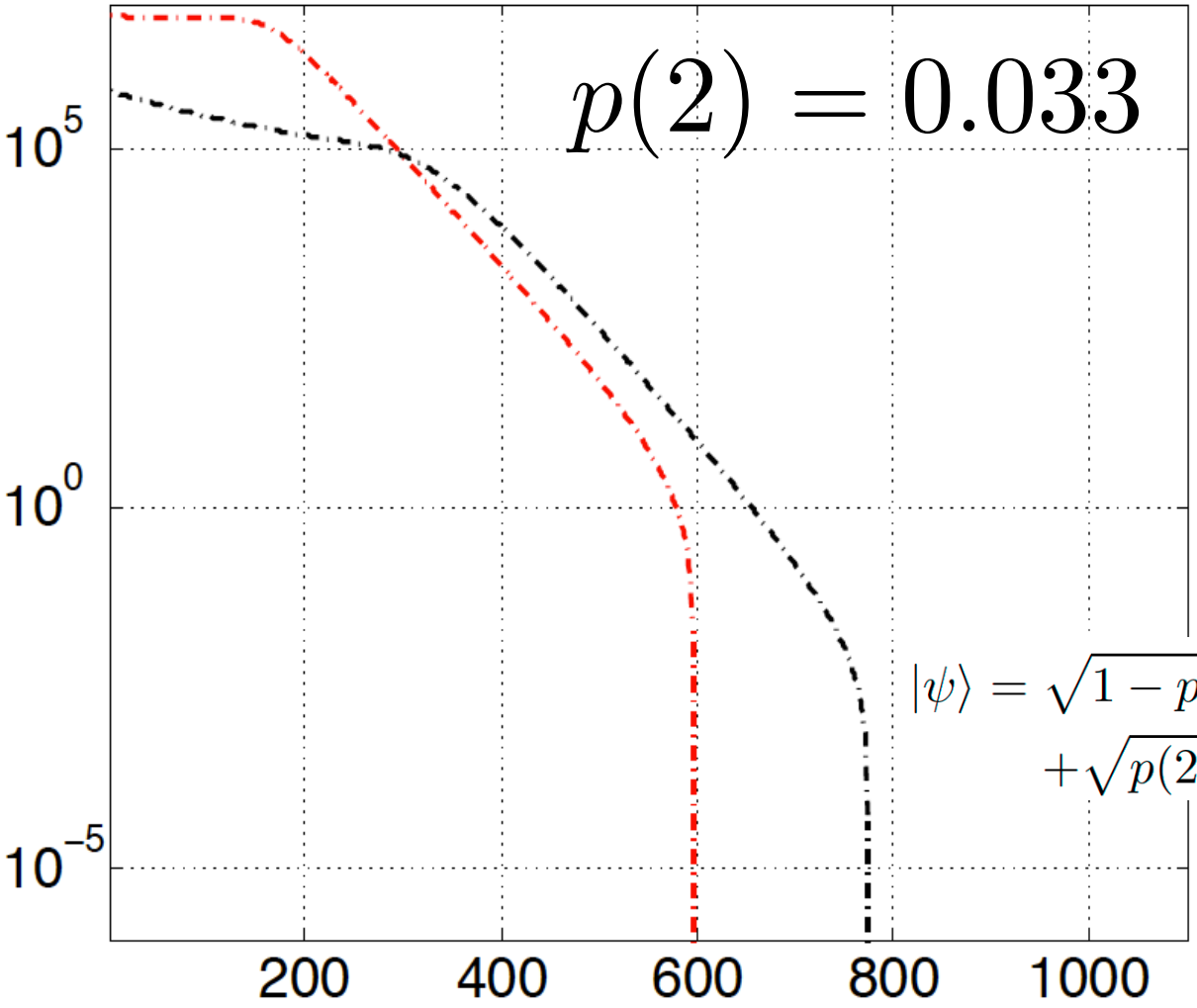
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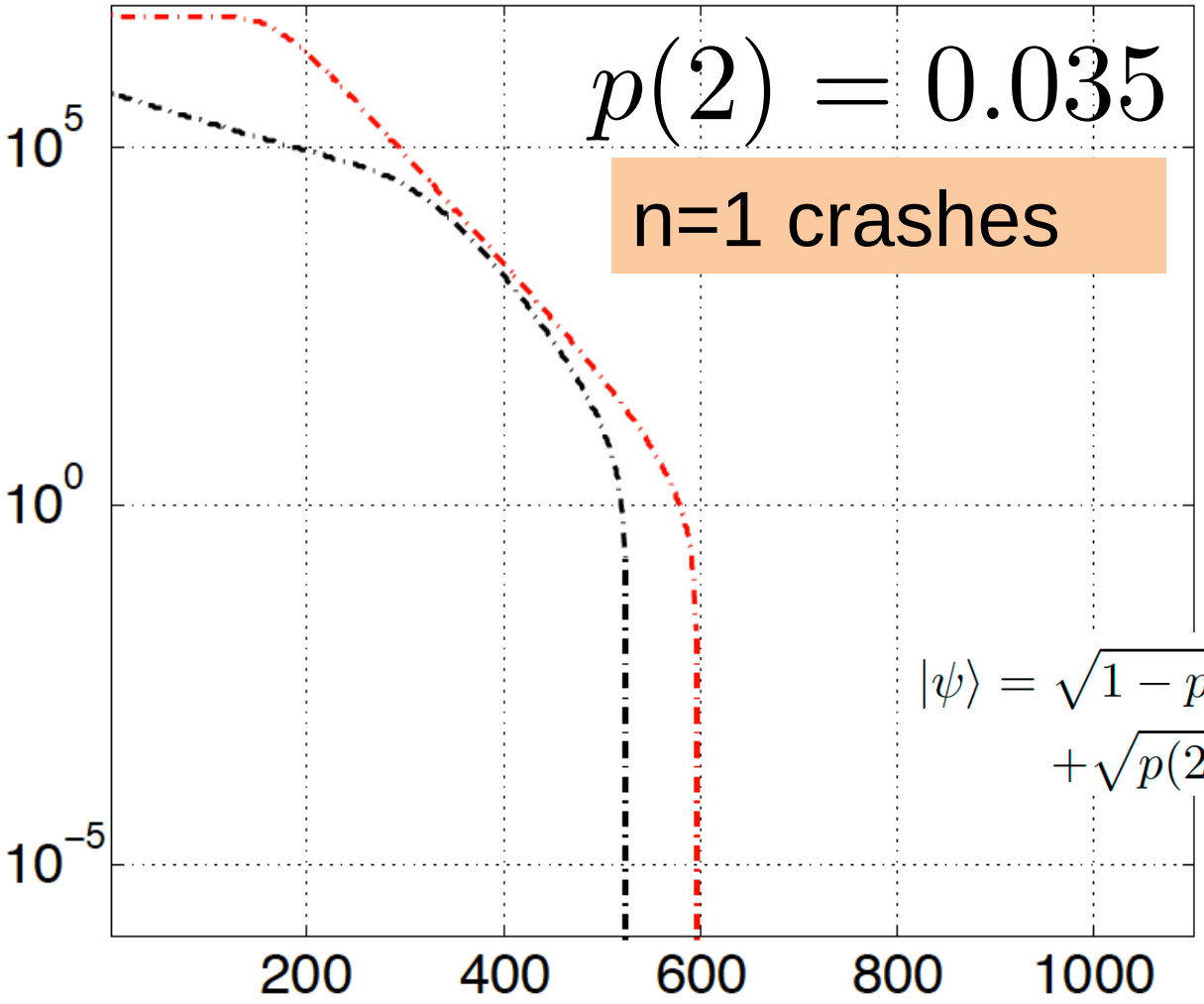
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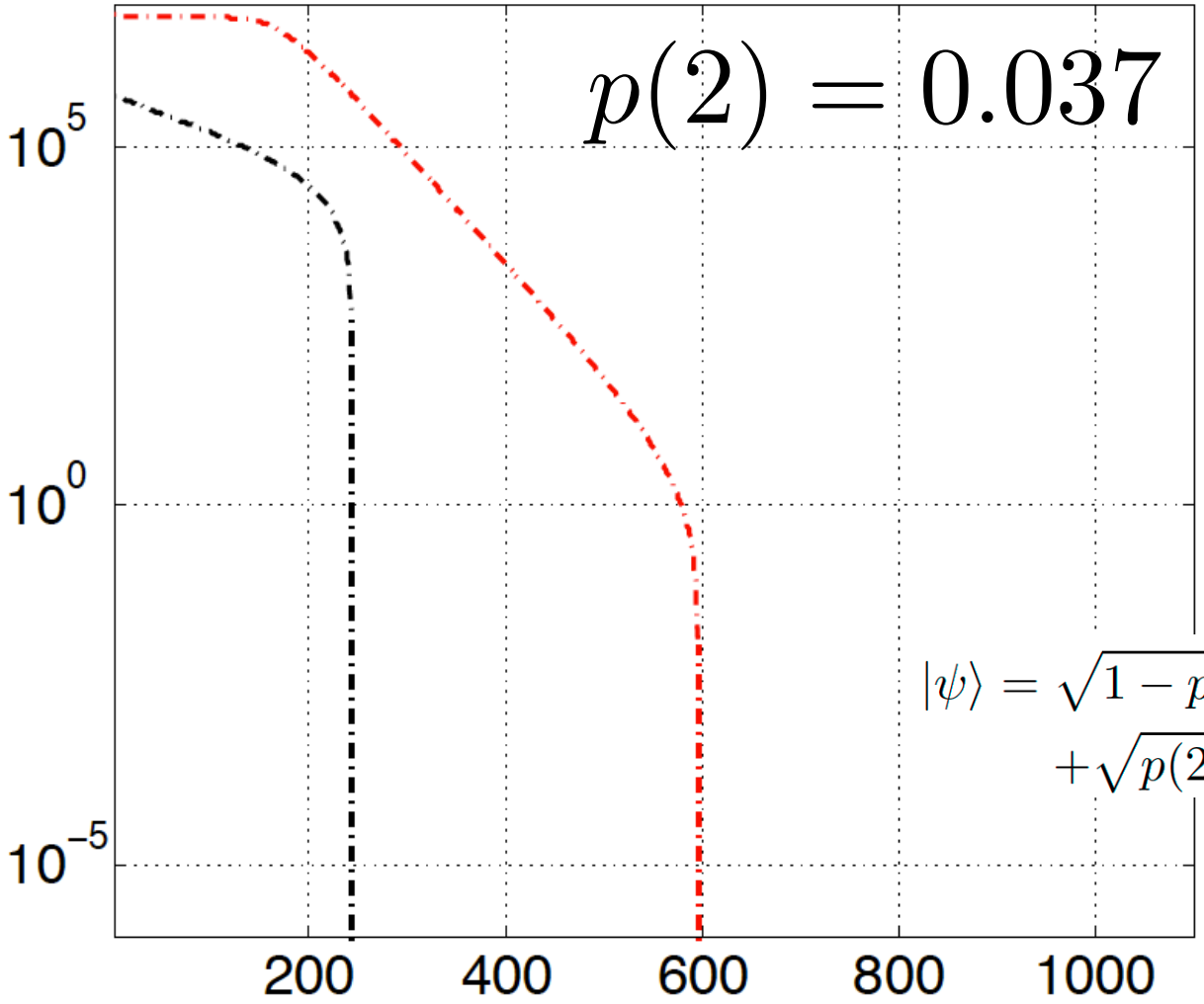
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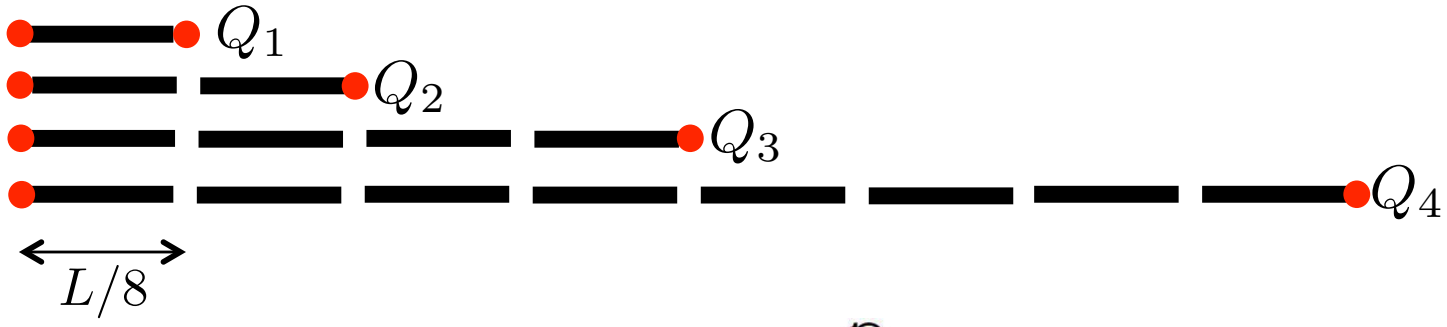
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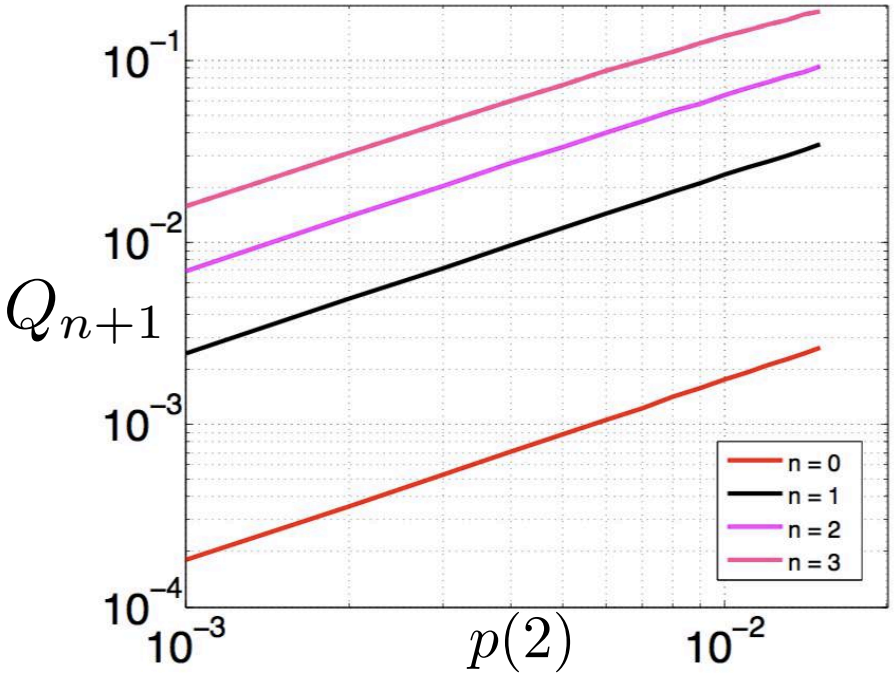
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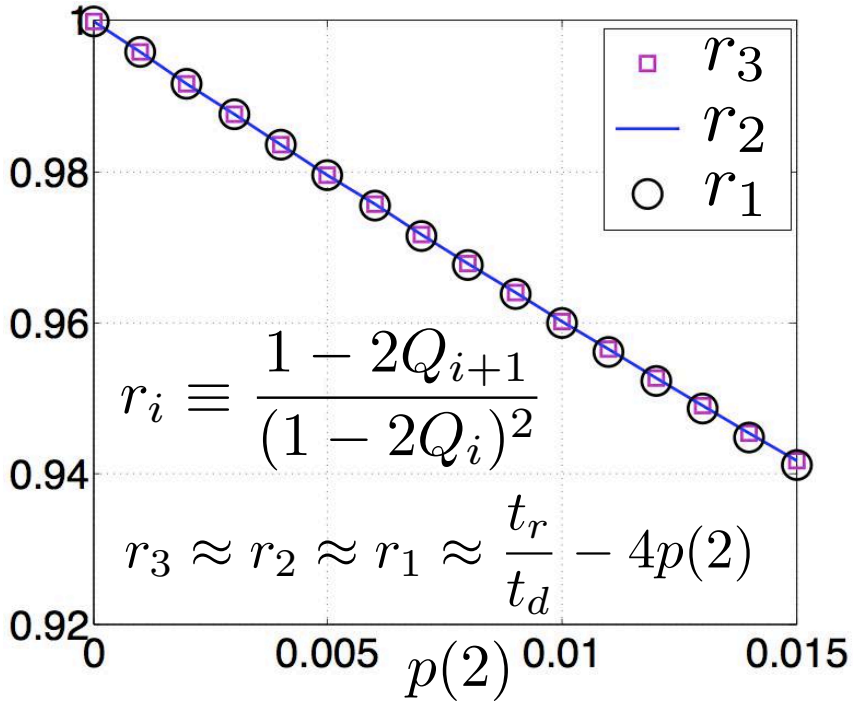
QBER propagation for non-ideal entanglement source

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- The QBER propagation law (from one swap level to next) holds even when p(2) > 0. Example:

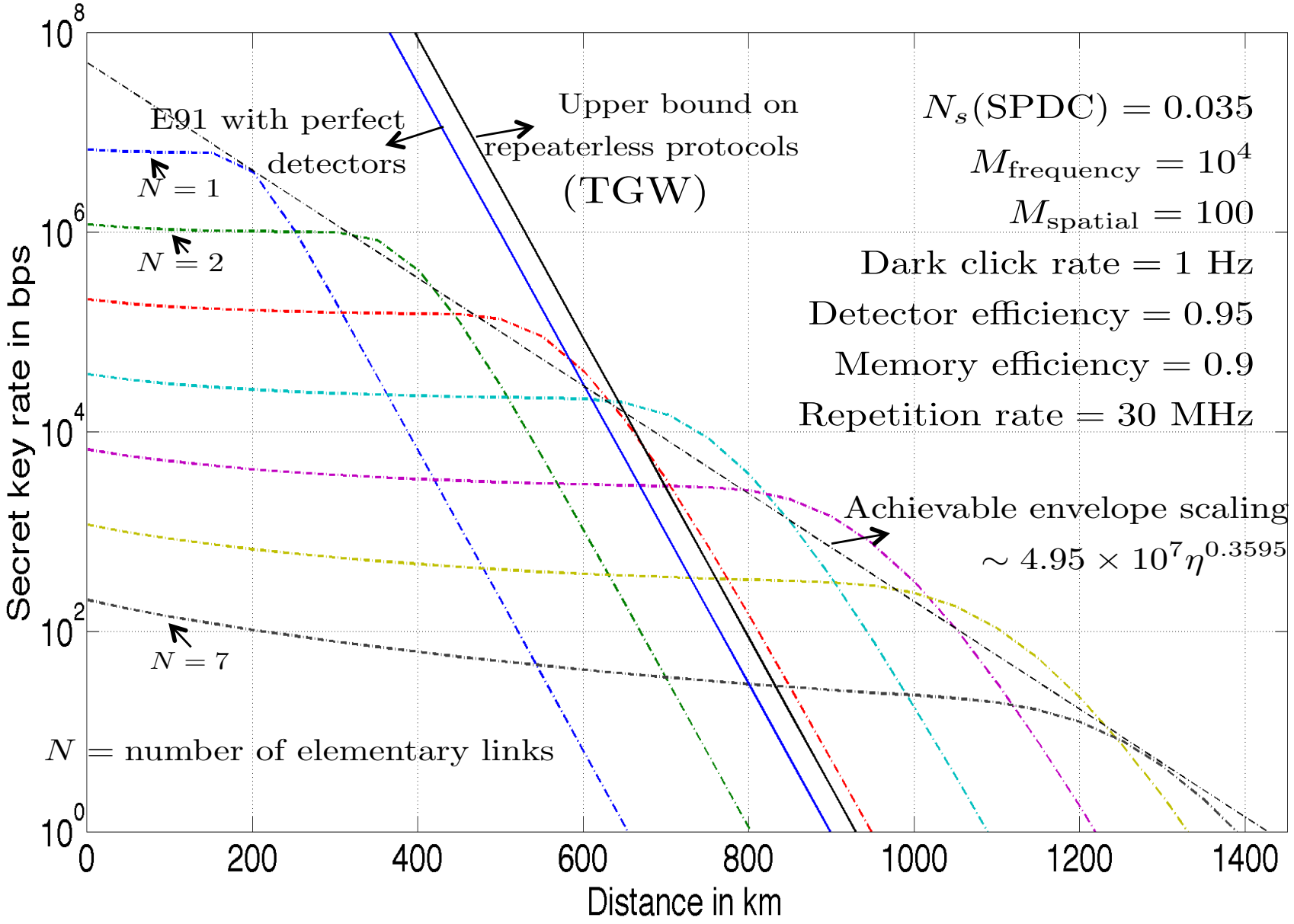






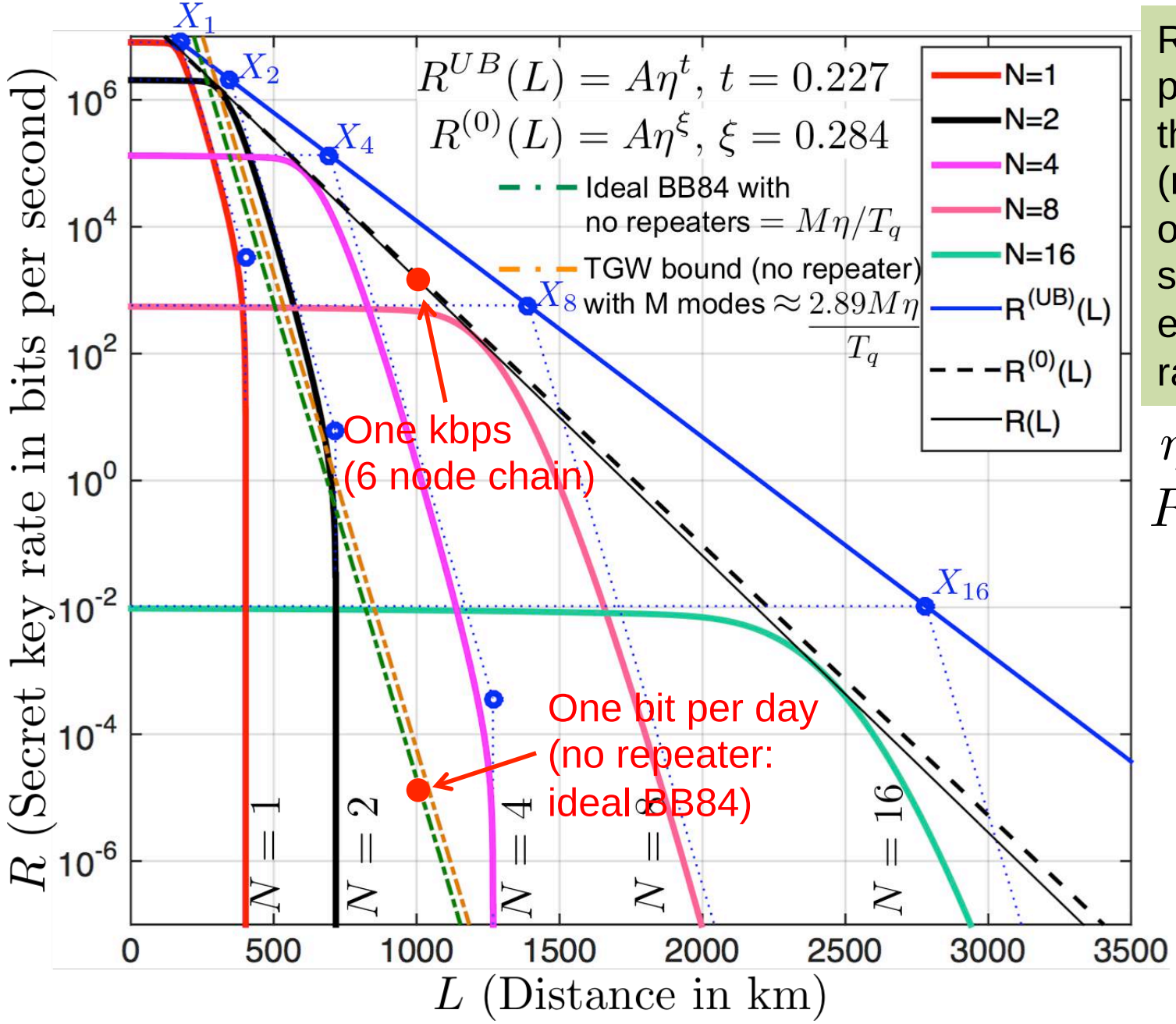
Rate vs loss envelope for SPDC and PNR architecture

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Rate-loss beats $R \propto \eta$, even using single "success" in each elementary link

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Rate-loss performance beats the "TGW" bound (repeater-less operation). But rate still decays exponentially with range

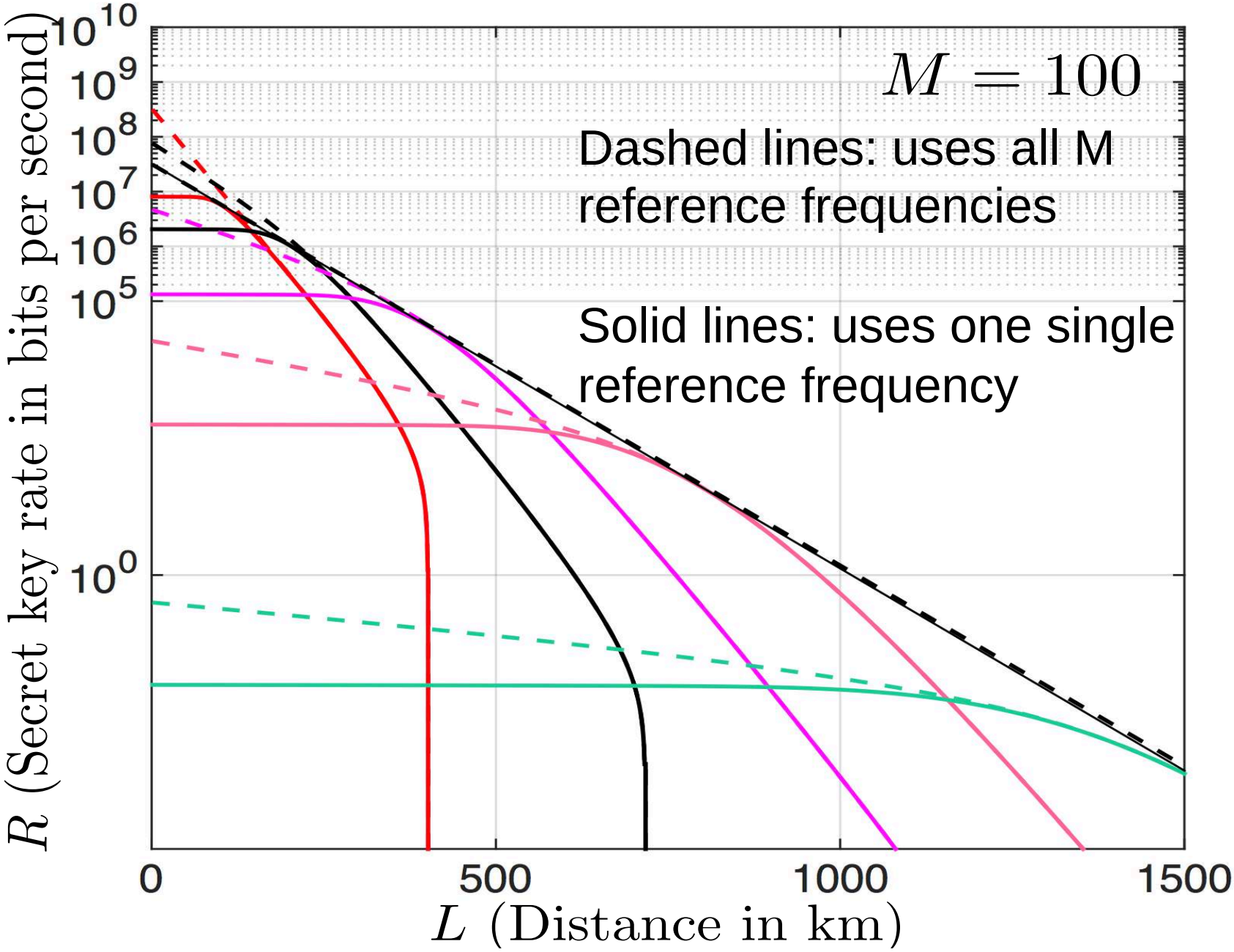
$$\eta = e^{-\alpha L}$$
$$R = Ae^{-\xi(\alpha L)}$$

Guha et al., arXiv: 1404.7183 (2014)
Guha et al., unpublished (2015)

P_d = 3E-5, η=0.9, M=1000 mode spectral multiplexing, λ_m=1dB, α=0.15 dB/km, T_q=50ns

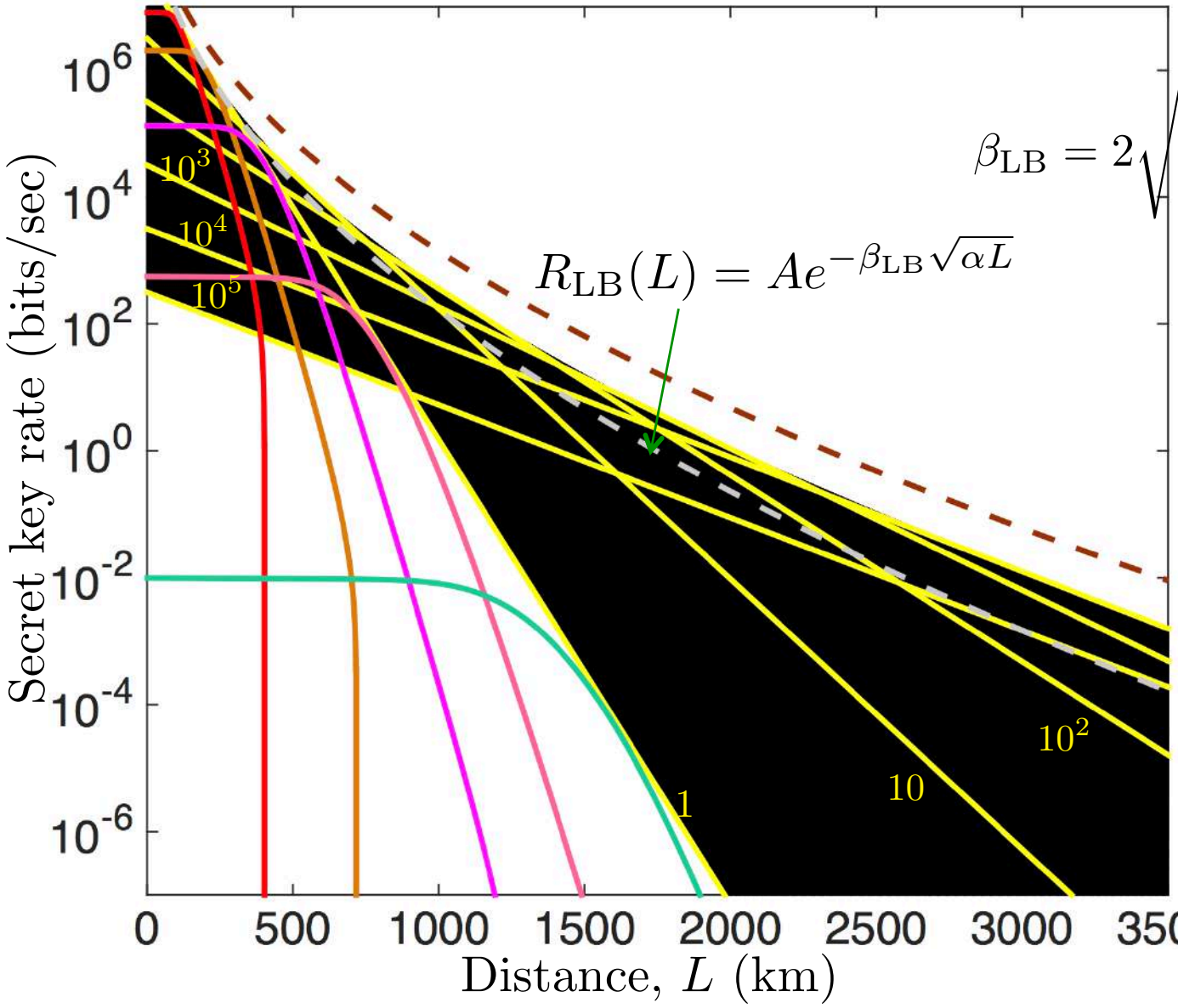
Using multiple successes within an elementary link improve low-range rate

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Using successes over time and frequency modes: sub-exponential rate-distance scaling

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$$A = \frac{M_f \eta_d^2 \eta_e^2}{2\eta_r^2 \lambda_m^2 T_q}$$

$$\beta_{\mathrm{LB}} = 2\sqrt{\ln\left(\frac{2}{\eta_r^2\lambda_m^2}\right) - \ln\left(1 - \frac{1}{e}\right)}$$

α = Fiber's loss coefficient

The numbers in yellow are the number of temporal modes, M_t

Switching loss = 0 dB

$$\eta_r = \eta_d = \eta_e = 0.9$$
$$T_q = 50\,\mathrm{ns}$$
$$\alpha = 0.15\,\mathrm{dB/km}$$
$$\lambda_m = 10^{-0.1\gamma_m},\ \gamma_m = 1(\mathrm{dB})$$
$$P_d = P_r = P_e = 3 \times 10^{-5}$$
$$M_f = 100$$

Conclusions

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- We found the exact quantum state after arbitrary number of swaps with losses and dark clicks (with deterministic sources).
- Two photon terms even with a small probability are very detrimental to the rate.
- SPDCs with single photon detectors do not give good rates.
- However, when used in conjunction with PNR detectors and highly multimode quantum memories, the rates can be comparable to those of deterministic sources.
- Even with lossy devices, one can achieve sub-exponential rate-distance tradeoff using multiplexing over frequency and time
- However, there is a threshold on the switching loss (in dB per switching stage) beyond which the advantage of using temporal modes for multiplexing goes away.
- In that regime, one goes back to the exponential rate-distance scaling (achievable via just multiplexing over frequency).

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